

LOOM 2226 — Technical Paper Series

Introduction, Scope, and Cross-Paper Conventions

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Status: Final reconciled series introduction / non-governing derivation companion

Canon compliance: Audited against Canon Baseline Manifest v2.2, CANON I v2.1.1, CANON II v2.2, CANON III v2.1.1, and their explicit governing dependency chain. Research extensions do not silently amend canon. **Governing authority:** Canon Baseline Manifest v2.2; Conceptual Physics Engineering MVP v1.3; promoted Loom WP-1-WP-5; promoted Metric WP-8-WP-11; Inherited Courier Integrated Propulsion Baseline v1.1; CANON I v2.1.1; CANON II v2.2; CANON III v2.1.1.

Abstract

The LOOM 2226 technical series is a conditional research program for a hard-science-fiction setting. It does not claim that contemporary physics predicts nonlocal transport, controllable spacetime embedding, or Mc-299m technology. It instead fixes two fictional physical premises and asks how much of the resulting world can be forced to obey ordinary mathematics, conservation, thermodynamics, engineering closure, network economics, and institutional causality.

The series is organized as a dependency chain.

Paper I: physical possibility → Paper II: engineering realization → Paper III: civilizational propagation

Paper I defines the fictional seam. Paper II prevents that seam from becoming arbitrary machinery. Paper III prevents the machinery from becoming arbitrary worldbuilding.

The governing speculative budget contains two and only two physical miracles:

M1: degenerate physical embedding

M2: controllable relational susceptibility

M1 states that a dynamically natural bounded physical domain may preserve its protected internal identity while occupying more than one gauge-inequivalent external embedding. M2 states that specially prepared matter may couple coherently to the relational degrees of freedom distinguishing those embeddings and exert finite control authority over them.

Everything else must be ordinary physics, a mathematical analogy, a measured constitutive closure, a derived consequence, a setting calibration, or an explicitly open problem.

1. What the series claims

The series claims only a conditional result:

Given M1 and M2, a coherent effective theory can be constructed in which identity-preserving relational translation, subluminal metric transport, ordinary rocket propulsion, finite engineering constraints, and a causally asynchronous interstellar civilization coexist without requiring free energy, free momentum, arbitrary destination selection, destructive copying, independent FTL communications, or an additional chronology miracle.

The series does **not** claim that modern physics predicts M1 or M2.

The real scientific literature is used in three narrower ways:

1. to identify legitimate mathematical structures;
 2. to retain constraints that a fictional extension should not casually discard;
 3. to expose where the fictional bridge begins.
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2. Common epistemic vocabulary

All three papers use the same six labels:

- **E — Established:** ordinary mathematics or accepted contemporary science.
- **A — Analogy:** a real formal structure used as disciplined mathematical ancestry.
- **P — Postulate:** an explicit fictional physical premise.
- **C — Constitutive closure / canon calibration:** an in-universe measured law or declared setting calibration.
- **D — Derived:** follows from stated assumptions and mathematics.
- **O — Open:** unresolved and forbidden from silently carrying downstream design load.

The distinction matters because an equation can be mathematically respectable while the physical coupling it describes remains fictional.

3. Foundational state

At effective level the controlled physical state is represented by

$$\mathcal{S} = (D, \mathcal{A}_I, E, q_R, \mathbf{Q}),$$

where:

- D is a dynamically autonomous domain;
- $\mathcal{A}_I$ is its protected identity algebra;
- E is its external relational embedding;
- q_R labels the accessible causal sector;
- $\mathbf{Q}$ contains ordinary conserved and state variables required for transport accounting.

M1 acts primarily on the distinction between $\mathcal{A}_I$ and E . M2 supplies finite controllability.

The low-energy matching rule is

$$\mathcal{J} \rightarrow 0 \quad \implies \quad \mathcal{C}_{\mu\nu}^R \rightarrow 0,$$

so ordinary GR/QFT/thermodynamic behavior must be recovered away from actively driven relational states.

4. Relational topography

Ordinary metric distance remains physically real, but it is not assumed to exhaust physical adjacency:

$$d_{\text{metric}}(A, B) \neq d_{\text{relational}}(A, B).$$

This produces two distinct mobility regimes from the same relational sector.

Smooth motion on a connected embedding branch motivates continuous subluminal Alcubierre-family metric transport:

connected branch motion $\rightarrow$ subluminal metric transport.

A nonperturbative transition between separated embedding basins motivates Loom translation:

finite-action branch transition $\rightarrow$ Loom re-embedding.

These are not separate miracles. They are two engineering regimes of M1/M2.

That statement is currently strongest at the **state-space and hardware level**, not yet at the constitutive-law level. The metric response and Loom transition response are separately calibrated first-generation laws. A deeper theory should recover both as limits of one relational functional; until then the correct claim is **shared relational sector and active medium, independently measured regime response**.

5. Causality and chronology

The technologically accessible M1 manifold is restricted to one causally admissible sector. Every controllable committed transition obeys

$$q_R = \text{constant}, \quad \Delta T_R > 0.$$

At the governing effective-theory level, the accessible operating sector is restricted this way. **This is not yet a derived consequence of M1.** It is a load-bearing causal admissibility condition carried by the current MVP and therefore an explicit theoretical IOU on the two-miracle architecture.

The deeper theoretical burden remains open: a complete microscopic model must show that the same parent theory that permits M1/M2 naturally selects a single stable causal orientation. If that orientation ultimately requires an independently adjustable chronology-censor mechanism, the project has spent a third miracle and the two-miracle claim fails.

6. Conservation

Neither M1 nor M2 grants free braking, momentum reset, or extractable work from a restored cycle.

For a Loom route,

$$\mathbf{Q}_B^{\text{nat}} = \mathcal{T}_{AB}(\mathbf{Q}_A).$$

If the desired arrival state differs,

$$\Delta \mathbf{Q} = \mathbf{Q}_B^{\text{desired}} - \mathbf{Q}_B^{\text{nat}},$$

the mismatch must be supplied by ordinary physical reservoirs or external exchange.

A fully restored cycle must satisfy

$$\Delta \mathbf{Q}_{\text{total}} = 0 \quad \Rightarrow \quad W_{\text{net}} \leq 0.$$

This is why a Loom vessel remains a rocket.

CANON II v2.2 adds the corresponding crew-rated engineering rule: once a Loom attempt enters its committed interval, certified domain membership and attachment state are fixed for that attempt. Mid-commit attachment, severance, docking-state, cargo-boundary, or other membership changes invalidate the solution and force abort/quench/fault handling rather than creating a momentum-exchange exploit.

7. Current engineering reference

The reconciled reference courier used by Papers II and III is:

Quantity	Current reference
Wet mass	~1,158.5 t
Working fluid/water	300 t
Normal reaction mass	250 t
Unified relational plant	~88 t
Active Mc-299m	10.0 kg
Active tiles	100 × 100 g
Distributed nodes	208
Shared reversible field bank	2 GJ
Cryoplant	150 kW-class at 20 K
Normal Loom boundary	~1,900 m ²
Extended boundary	~2,430 m ²
Effective boundary patches	~971
Boundary authority	~670
Loom target action	16
Attempt frequency	10 ⁹ s ⁻¹
Commit interval	0.25 s
Selectivity separation	40
Leakage target	10 ⁻¹² per committed gate
Metric NORMAL	0.268c
Metric FAST	0.437c
Metric EXPEDITE	0.519c
Metric HARD	0.595c
Metric saturation response	~0.678c, uncertified
Metric FTL	hard kill

Legacy 32 kg Mc sizing, 64 × 0.5 kg cells, separate legacy metric packages, area-times- v^2 sizing, and the historical 0.03–0.05c card are non-governing.

8. Engineering hierarchy

Paper II preserves three mobility regimes:

$$\boxed{\text{torch} + \text{sub-c metric} + \text{Loom}}$$

with distinct jobs.

Torch handles local momentum exchange, docking, braking, rescue, combat maneuver, orbital mechanics, and arrival-state correction.

Metric handles continuous long-baseline Solar transport in characterized space, with large terminal stand-off and environmental certification.

Loom handles discontinuous route-limited relational translation.

The doctrine can be summarized as:

Torch moves mass. Metric moves urgency. Loom moves the frontier.

9. Current civilizational reference

Paper III uses the current demographic/economic control rather than superseded population assumptions.

Key working values are:

Quantity	Current status
Conserved biological humanity	8,571,824,600 - canon
Recognized synthetic persons	214,068,429 - canon
Biological humans off Earth	~259.3M, conserved within biological total - canon/derived
Major/strategic Solar infrastructure graph	127 nodes - canon
Primary flight-grade Mc production	~1,000 kg/y - canon
Recovered/requalified Mc return flow	~150-250 kg/y - canon
Working central serviceable Mc flow	~1.2 t/y - canon calibration
Cumulative flight-grade Mc	~8.7 t - canon
Qualified Mc demand	modestly exceeds normal serviceable supply at the margin - canon
Continuous civilization electricity	~66.8 TW - Paper III working model output, not promoted canon
Independent FTL communications	none - canon
Interstellar demographic effect by 2226	Loom too recent to drive mature Solar demography - canon constraint

The promoted Solar census is already conserving. Off-Earth biological residents are part of the same 8,571,824,600 biological-human ledger. Recognized synthetic persons total 214,068,429 and are counted separately from non-person automation.

10. Two geographies and packet civilization

Paper III formalizes the discovered Loom network as a weighted graph

$$G = (V, E),$$

with effective economic distance

$$d_G(i, j) = \min_{p:i \rightarrow j} \sum_{e \in p} c_e.$$

Therefore a star can be closer in light-years and farther in operational relational cost.

Because there is no independent FTL communications system, information must be physically carried. Information state therefore has causal age:

$$a_I = t_{\text{local}} - t_{\text{source}}.$$

This creates packet markets, delayed authority, stale contracts, route-sensitive insurance, local delegated power, and a frontier whose constitutional problems are consequences of transport physics rather than decorative lore.

11. High Strangeness firewall

The relational framework permits a bounded research question about anomalous experience, but it does not convert anecdote into calibration evidence.

A passive system may possess weak incidental responsivity

$$\chi_R^{\text{inc}} \neq 0, \quad \mathcal{J} = 0,$$

meaning it may respond weakly to a genuine relational disturbance without possessing useful back-reaction or control authority.

The valid inference direction is

independently instrumented anomaly $\rightarrow$ test for physical or cognitive correlation

not

strange experience $\rightarrow$ assume relational cause.

Relational topography can therefore provide a fictional framework in which a small residual of synchronicity, anomalous perception, UAP-like observations, apparitional reports, psychedelic correlations, or other high-strangeness phenomena might someday prove physically interesting. Most such experiences remain ordinary, mistaken, fraudulent, psychological, technological, statistical, or unresolved.

The governing firewall is:

hidden adjacency $\neq$ destiny.

Human agency is not replaced by topology.

12. Red-team revision status

The August 2026 graduate red-team pass independently reproduced the principal Loom probability, rocket-equation, radiator, torch-deposition, and population-ledger arithmetic. It also exposed three paper-level closure defects now corrected in this series revision: the metric ramp-distance derivation, the cryoplant electrical conversion chain, and the missing full-array authority reconciliation. Chronology orientation remains deliberately open rather than being papered over by definition.

12. Shared hard kills

Across the series, the current architecture fails if it requires:

- a third independent miracle;
- FTL metric transport;
- free energy, momentum, or braking;

- matter duplication;
- destructive replacement masquerading as continuity;
- controllable closed causal loops;
- consciousness control of M2;
- arbitrary narrative modification of the hidden route graph;
- anecdotal calibration of physical constants;
- ship performance without mass/power/heat closure;
- unsupported bulk interstellar economics;
- retroactive Solar demographics caused by a technology only a few years old;
- remote real-time institutional knowledge without a causal information channel.

A hard kill is not patched with prose. The affected machine, setting assumption, or miracle budget must change.

13. Open research program

The most consequential unresolved targets remain:

1. microscopic origin and large-domain scaling of M1/M2;
 2. natural emergence of bounded dynamically autonomous domains;
 3. causal-sector orientation without hidden M3;
 4. Mc-299m production physics and aging;
 5. M2 constitutive law beyond point calibration;
 6. full four-dimensional metric RSET benchmark;
 7. metric geodesic/caustic and clustered-node certification;
 8. torch plasma/radiation/nozzle deposition closure;
 9. globally conserving Solar-System demographic engine;
 10. habitat-capital refinement and synthetic-person individuation/staffing calibration;
 11. 2226 input-output and ownership models;
 12. route-flow, insurance, and port-throughput models;
 13. topology spectroscopy and mis-translation mechanics;
 14. High Strangeness mechanics that preserve uncertainty and skepticism.
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14. Series map

The four documents should be read in this order:

Introduction — Scope and Cross-Paper Conventions

Defines the common vocabulary, governing numerical card, supersession rules, and boundaries between canon, derivation, and open research.

Paper I — Mathematical Foundations of Relational Embedding

Asks whether M1/M2 can be stated as a constrained effective theory without collapsing into coordinate freedom, arbitrary magic, destructive copying, free energy, or chronology handwaving.

Paper II — Engineering Consequences of Relational Transport

Asks what a ship must physically contain, reject, measure, certify, and survive if Paper I is granted.

Paper III — Economic Geography and Civilizational Dynamics of Relational Transport

Asks what those machines do to prices, networks, geography, institutions, information, demography, and political authority.

Together:

physics → engineering → economics → institutions → human stories.

That dependency direction is the governing discipline of LOOM 2226.

LOOM 2226 — Paper I

Mathematical Foundations of Relational Embedding

A minimal two-postulate effective framework for identity-preserving nonlocal translation and subluminal metric transport

Version: 1.0

Date: 20 August 2026

Status: Technical research paper / foundational derivation companion

Canon compliance: Audited against Canon Baseline Manifest v2.2, CANON I v2.1.1, CANON II v2.2, CANON III v2.1.1, and their explicit governing dependency chain. Research extensions do not silently amend canon.

Governing project basis: Canon Baseline Manifest v2.2; Conceptual Physics Engineering MVP v1.3; promoted Loom WP-1-WP-5; promoted Metric WP-8-WP-11; CANON II v2.2; Relational Embedding Theory Graduate Second-Round Review v0.2.

This paper does not claim that contemporary physics predicts LOOM technology.

Abstract

LOOM 2226 adopts a deliberately narrow speculative budget. Ordinary general relativity, quantum field theory, quantum information, thermodynamics, and conservation laws are retained as infrared matching requirements. Two fictional postulates are added: **M1, degenerate physical embedding**, which permits a dynamically natural bounded subsystem to possess multiple gauge-inequivalent external embeddings while preserving a protected internal observable algebra; and **M2, controllable relational susceptibility**, which permits specially prepared matter to couple coherently to the relational degrees of freedom distinguishing those embeddings. This paper develops the mathematical foundations of that framework without claiming a complete microscopic theory of quantum gravity.

A pregeometric parent-action candidate is introduced as a research scaffold rather than canon. Its semiclassical ordered phase supplies a moduli space of external embeddings. Smooth motion on a connected branch yields continuous reconstructed displacement and motivates the promoted subluminal metric mode; non-perturbative transitions between separated minima yield Loom translation. Protected observable algebras provide an operational identity criterion; dynamical autonomy and algebraic leakage provide a non-geometric definition of the translated domain; global composability replaces an ad hoc paradox selector; Noether/Ward structure and explicit reservoirs enforce conservation; and a single accessible causal sector with monotone relational ordering prevents controllable closed causal loops.

The paper distinguishes established mathematics, analogy, explicit fictional postulate, effective constitutive closure, and open problem. The purpose is not to disguise fiction as discovery. It is to make the fictional seam narrow enough that everything downstream can be attacked with ordinary scientific discipline.

1. Epistemic standard

The framework uses six labels:

- **E — Established:** ordinary mathematics or accepted physical theory.
- **A — Analogy:** a real formal structure used as a disciplined analogy.
- **P — Postulate:** an explicit fictional physical premise.
- **C — Constitutive closure:** an in-universe empirical law or calibrated response.
- **D — Derived:** follows from stated assumptions and mathematics.
- **O — Open:** unresolved and forbidden from silently doing engineering work.

The core rule is simple: **a citation beside a fictional equation does not mean the cited literature predicts LOOM.** It means the mathematical move has a legitimate scientific ancestry.

2. Stopping rule and ordinary-physics matching

LOOM does not attempt to derive the Standard Model, solve quantum gravity, or identify the ultimate ontology of spacetime. Instead it imposes the infrared matching condition

$$\mathcal{J} \rightarrow 0 \implies \mathcal{C}_{\mu\nu}^R \rightarrow 0,$$

where $\mathcal{J}$ is engineered relational control authority and $\mathcal{C}_{\mu\nu}^R$ denotes any effective relational correction to ordinary spacetime dynamics.

Thus, away from deliberately driven relational states, the theory must recover ordinary local physics to experimental accuracy:

$$\text{LOOM IR} \rightarrow \text{GR, QFT, SM phenomenology, thermodynamics.}$$

This is a matching requirement, not a claimed derivation.

3. Engineering state and equivalence structure

At effective level define a controlled physical state

$$\mathcal{S} = (D, \mathcal{A}_I, E, q_R, \mathbf{Q}),$$

where (D) is a dynamically autonomous domain, $\mathcal{A}_I$ its protected identity algebra, (E) its external relational embedding, q_R its accessible causal-sector label, and $\mathbf{Q}$ the ordinary conserved/state variables needed for arrival-state accounting.

The separation between $\mathcal{A}_I$ and (E) is essential. LOOM does not identify an object with its coordinate location. Nor does it identify all changes of (E) as physical. Gauge-related descriptions are quotiented out.

4. Postulate M1 — degenerate physical embedding

For a dynamically natural domain (D), there exist at least some global configurations C_A, C_B such that

$$\mathcal{G}(C_A)|_D \simeq \mathcal{G}(C_B)|_D,$$

while

$$\mathcal{G}(C_A)|_{\bar{D}} \not\simeq \mathcal{G}(C_B)|_{\bar{D}}.$$

after quotienting ordinary gauge redundancy.

Equivalently, for fixed protected identity algebra,

$$\mathcal{M}_E(\mathcal{A}_I) = \{[E] : \mathcal{G}(E)|_D \simeq \mathcal{A}_I\} / \text{gauge}.$$

contains more than one physically inequivalent point.

Status: P.

M1 is not coordinate freedom. It asserts multiple physical global solutions with the same protected internal structure and different exterior reconstruction. It is also not ordinary purification non-uniqueness, copying, destructive reconstruction, a wormhole, a hidden corridor, or travel through a second spatial manifold.

The immediate consequence is relational topography:

$$d_{\text{metric}}(A, B) \not\equiv d_{\text{relational}}(A, B).$$

Ordinary astronomical distance remains physically real; it simply does not exhaust the deeper adjacency structure.

5. Why M1 is not justified by emergent-spacetime research

Modern quantum-gravity research contains serious programs in which spacetime is emergent from non-spatiotemporal quantum degrees of freedom. Group-field-theory condensate cosmology is one explicit example in which collective states of more fundamental degrees of freedom are used to reconstruct effective cosmological geometry [1,2]. These programs establish that “spacetime may be emergent” is scientifically legitimate research territory.

They do **not** establish M1.

LOOM borrows only the methodological permission to distinguish a deeper relational state from its effective spacetime reconstruction. The existence of macroscopic, identity-preserving, gauge-inequivalent re-embeddings is the first fictional miracle.

6. Postulate M2 — controllable relational susceptibility

Let ϕ denote the primitive relational degrees of freedom of a candidate parent theory. M2 states that specially prepared matter can generate a controllable source J_A coupled to genuine relational observables $\mathcal{O}_R^A$:

$$S_R \rightarrow S_R + S_{\text{ctrl}}, \quad S_{\text{ctrl}} = \int J_A, \mathcal{O}_R^A.$$

Status: P.

M2 must cover both continuous motion on an embedding branch and nonperturbative transition between branches. It supplies no separate warp force, teleportation force, negative-energy exemption, free momentum, inertial damper, chronology censor, or consciousness field.

At engineering level M2 is summarized by the measured chain

$$(N_a, W_a, T) \rightarrow \mathcal{J} \rightarrow B_j \rightarrow \Gamma_j \rightarrow P_j.$$

The microscopic origin of $\mathcal{J}$ is open; its engineering response is calibrated.

7. Candidate parent action

A research-level parent candidate can be written without primitive spacetime coordinates:

$$S_R[\phi] = \frac{1}{2} \langle \phi, \mathcal{K} \phi \rangle + \sum_{n=3}^{n_{\max}} \frac{\lambda_n}{n!}, \mathcal{V}_n[\phi^{\otimes n}].$$

The corresponding quantum valuation is

$$Z = \int \mathcal{D}\phi, e^{iS_R[\phi]/\hbar}.$$

Status: A/O.

This resembles the architecture of background-independent field-theoretic approaches such as group field theory, where combinatorial or group-valued degrees of freedom and interaction/gluing kernels can precede continuum spacetime [1,2]. It is not asserted that the LOOM universe literally is a known GFT.

The candidate earns explanatory value only if the kernels do not merely encode the desired macroscopic world by hand.

For a saddle ϕ_0 ,

$$\left. \frac{\delta S_R}{\delta \phi} \right|_{\phi_0} = 0,$$

and the fluctuation operator is

$$H_{ab} = \left. \frac{\delta^2 S_R}{\delta \phi_a \delta \phi_b} \right|_{\phi_0}.$$

A stable ordered phase requires an acceptable spectrum of (H); elegance without a stable fluctuation spectrum is not a solution.

8. Global composability

Earlier versions used an explicit admissibility selector. Paper I removes that device.

For boundary relational data ψ_Σ , define

$$\mathcal{A}[\psi_\Sigma] = \int * \phi \| * \Sigma = \psi_- \Sigma \mathcal{D} \phi, e^{iS_R/\hbar}.$$

If no globally composable configuration satisfies the proposed constraints,

$$\mathcal{C}(\psi_\Sigma) = \emptyset,$$

then

$$\mathcal{A}[\psi_\Sigma] = 0.$$

Status: A/D conditional on the parent theory.

This eliminates logically inconsistent assignments without invoking an external paradox detector. It does **not** by itself eliminate globally self-consistent closed causal curves; chronology requires separate analysis.

9. Emergent ordered phase and relational time

Suppose the parent action admits an ordered phase with schematic condensate

$$\langle \phi \rangle = \rho_0 e^{-i\mu T_R}.$$

Here T_R is an emergent relational clock coordinate of the ordered phase, not primitive Newtonian time.

The phase also selects an orientation/charge branch,

$$Q_R > 0.$$

LOOM restricts technologically accessible embeddings to one connected causal sector:

$$q_R = \text{constant}, \quad \Delta T_R > 0$$

for every controllable committed transition.

Status: P/O at the governing MVP level; not yet derived from M1/M2.

The current effective theory restricts certified operation to this sector, but that restriction is an **undischarged theoretical obligation**, not a solved consequence of degenerate embedding. The key academic issue is whether the same ordered phase that reconstructs ordinary spacetime naturally selects one stable orientation

while leaving the low-energy matter sector Lorentz invariant to observed precision. If a separately adjustable veto must be imposed solely to stop time machines, that veto is a hidden third miracle and the two-miracle reduction fails.

The need for caution is real. Hawking’s chronology-protection work shows why closed timelike curves and chronology horizons are not a trivial side issue in relativistic theories [3]. LOOM does not claim Hawking proves $\Delta T_R > 0$; the literature establishes the seriousness of the constraint.

10. Low-energy causal speed

Within the reconstructed phase, define the ordinary low-energy causal speed operationally from a massless dispersion relation:

$$c^2 = \lim_{k \rightarrow 0} \frac{\partial \omega^2}{\partial k^2}.$$

Approximate Lorentz recovery requires ordinary sectors to share this limiting cone to empirical precision.

The crucial conceptual separation is therefore

$$c = \text{local causal propagation speed of the IR phase,}$$

not

$$c = \text{maximum rate of change of every deeper relational degree of freedom.}$$

That second inequality is not available in standard physics; M1 supplies the fictional exception.

11. Protected identity algebra

For domain (D), let $\mathcal{A}_D$ be its observable algebra and let

$$\mathcal{A}_I \subseteq \mathcal{A}_D$$

be the subset designated as identity-bearing for the physical task.

An ideal Loom channel $\mathcal{N}_L$ preserves identity when

$$\mathcal{N}_L^\dagger(O) = O \quad \forall O \in \mathcal{A}_I.$$

For an imperfect device define

$$\epsilon_I = \sup_{|O| \leq 1} *_{O \in \mathcal{A}_I} |\mathcal{N}_L^\dagger(O) - O|.$$

Certification requires

$$\epsilon_I \leq \epsilon_{I,\max}.$$

Status: A/C.

This formalism has genuine quantum-information ancestry. Knill and Laflamme derived conditions for perfect recovery of encoded quantum information [4]. Blume-Kohout et al. showed that information-preserving structures of quantum processes are naturally described by matrix algebras and fixed observables [5]. LOOM uses that algebraic style operationally; it does not claim to have solved philosophical personal identity.

12. No duplication

A successful transition may change external embedding while preserving $\mathcal{A}_I$, but it may not yield two independently usable output domains carrying the same arbitrary protected quantum information.

The rule is stronger and more macroscopic than the quantum no-cloning theorem, but it is conservative with respect to ordinary quantum mechanics. Wootters and Zurek showed that linear quantum mechanics forbids perfect cloning of an arbitrary unknown quantum state [6].

Thus LOOM's duplication prohibition is not proof-derived from no-cloning; it is a foundational firewall consistent with it.

13. Dynamical autonomy and the domain boundary

A translated domain is not an arbitrary geometric volume. Define a conditional expectation/projection

$$\mathbb{E}_D : \mathcal{A} \rightarrow \mathcal{A}_D$$

and an effective generator $\mathcal{L}^\dagger$. Define leakage

$$\epsilon_{\text{leak}}(D) = \sup_{|O| \leq 1}^{O \in \mathcal{A}_D} |(1 - \mathbb{E}_D)\mathcal{L}^\dagger(O)|.$$

A useful domain has small leakage over the relevant control interval.

Under engineered boundary control,

$$\mathcal{L} = \mathcal{L}_D + \mathcal{L}_{\bar{D}} + s_B \mathcal{L}_{\text{cross}}, \quad s_B \ll 1,$$

so that

$$\epsilon_{\text{leak}}(D) \downarrow.$$

Status: A/C.

This definition is intentionally dynamical. A docked craft, cable, radiator panel, fluid plume, or partially detached cargo can change domain membership because it changes coupling, not because a game master redraws a bubble.

If competing domain cuts differ by Euclidean/control action ΔS_B , a useful semiclassical selectivity scale is

$$\frac{\Delta S_B}{\hbar} \gtrsim \ln \left(\frac{N_B}{p_B} \right).$$

For $N_B = 10^6$ serious competing cuts and $p_B = 10^{-12}$,

$$\Delta S_B/\hbar \gtrsim 41.4.$$

This is a stress-test scale, not a prediction.

14. Embedding moduli

On a smooth connected branch of $\mathcal{M}_E$, let h^a be local coordinates. The low-energy effective Lagrangian can be written

$$L_E = \frac{1}{2} G_{ab}(h) \dot{h}^a \dot{h}^b - V_E(h) + J_a h^a.$$

The effective moduli-space metric is

$$G_{ab} = \frac{\partial^2 L_E}{\partial \dot{h}^a \partial \dot{h}^b}.$$

The reconstructed spacetime and domain position depend on the embedding:

$$g_{\mu\nu} = g_{\mu\nu}(h), \quad X^\mu = X^\mu(h).$$

Hence

$$v^i = \frac{\partial X^i}{\partial h^a} \frac{dh^a}{dT_R}.$$

Status: A/F2.

This is the cleanest mathematical unification of metric and Loom behavior: metric mode is smooth motion on a connected branch; Loom is a nonperturbative jump between separated basins.

15. Continuous metric transport as a smooth relational branch

Canon-interface note. This is a mathematical interpretation, not the withdrawn v2.0 “phase-walk” propulsion architecture. Governing engineering is Metric WP-8–WP-11: sub-c Alcubierre-family geometry, measured susceptibility, certified speed card, null-geodesic certification, and FTL hard kill.

Choose a local branch coordinate β whose reconstructed metric is approximately

$$ds^2 = -c^2 dT^2 + [dx - c\beta f(r) dT]^2 + dy^2 + dz^2.$$

At the controlled center,

$$v_M = c\beta.$$

This is a coordinate definition within the effective branch, not a universal microscopic law.

Expand the effective potential:

$$V_M(\beta) = \frac{1}{2}\kappa_2\beta^2 + \frac{1}{4}\kappa_4\beta^4 + \dots,$$

with local stability requiring $\kappa_2 > 0$. The source needed to hold the state is

$$J_\beta = \frac{dV_M}{d\beta} = \kappa_2\beta + \kappa_4\beta^3 + \dots.$$

At weak bias, field energy is generically quadratic:

$$E_M \sim \frac{A}{\delta} \left(\frac{1}{2} K_2 \beta^2 + \dots \right).$$

This explains why low-speed power or stored-energy scalings can begin as v^2 without elevating that scaling to a fundamental law.

16. Relation to real warp-drive mathematics

Alcubierre-type metrics demonstrate that GR admits transport geometries with nontrivial shift structure. Bobrick and Martire developed a broader physical classification and emphasized that warp shells do not

grant free self-acceleration [7]. However, standard-GR warp constructions face severe stress-energy constraints. Santiago, Schuster and Visser showed that physically reasonable generic warp drives violate standard energy conditions within the frameworks they analyzed [8].

LOOM therefore makes a precise separation:

$$\text{GR literature} \Rightarrow \text{geometry} + \text{diagnostics} + \text{failure tests},$$

while

$$\text{M1/M2} \Rightarrow \text{fictional constitutive access to embedding degrees}.$$

The promoted LOOM metric branch is deliberately subluminal. FTL metric remains a hard kill. This reduces, but does not eliminate, the need for geodesic, horizon, energy-condition, and semiclassical-QFT scrutiny.

17. Nonperturbative Loom transition

Let E_A and E_B correspond to separated minima or basins in the effective embedding landscape. A Loom event is modeled semiclassically by an instanton/bounce-like path in Euclideanized configuration space.

Define an effective Euclidean action

$$S_E[h] = \int d\tau \left[\frac{1}{2} G_{ab}(h) \frac{dh^a}{d\tau} \frac{dh^b}{d\tau} + V_E(h) - J_a h^a \right].$$

For candidate endpoint (j), define the dimensionless transition action

$$B_j = \frac{S_{E,j}}{\hbar}.$$

Then the semiclassical transition rate has generic form

$$\Gamma_j = \Omega_j e^{-B_j}.$$

Status: A/C.

The exponential is scientifically well motivated by semiclassical tunneling and activated-process mathematics; the relational action itself is fictional.

At engineering level the adopted target-bias closure is

$$B_j = \max[0, B_j^0 - M_j \mathcal{J}].$$

This linear action reduction is not fundamental. It is a first-generation constitutive law to be replaced if experiment shows nonlinear response.

18. Selectivity

For target (T) and competitor (j),

$$\frac{\Gamma_j}{\Gamma_T} = \frac{\Omega_j}{\Omega_T} e^{-(B_j - B_T)}.$$

If prefactors are comparable and there are N_c serious competitors, requiring total mis-selection probability below p_{mis} gives the useful scale

$$\Delta B_{\text{select}} \gtrsim \ln \left(\frac{N_c}{p_{\text{mis}}} \right).$$

An action separation of 40 produces

$$e^{-40} \approx 4.25 \times 10^{-18}.$$

Thus very high selectivity can arise from finite action margins rather than infinitely precise coordinate targeting.

19. Markovian engineering regime

If decoherence and bath coupling erase coherent multi-route phase information rapidly compared with the gate interval, the first-generation machine can be treated as competing Poisson channels:

$$\Gamma_{\Sigma} = \sum_j \Gamma_j,$$

$$P_j = (1 - e^{-\Gamma_{\Sigma} \tau_g}) \frac{\Gamma_j}{\Gamma_{\Sigma}},$$

$$P_0 = e^{-\Gamma_{\Sigma} \tau_g}.$$

Status: A/C.

This is ordinary rate-process mathematics. It is valid only inside the measured Markovian envelope. Observation of memory, interference, or history-dependent rates requires a richer open-system model rather than narrative hand-waving.

20. Conservation from parent symmetry

If the parent action is invariant under a continuous transformation, the corresponding Ward/Noether structure constrains conserved quantities. In the reconstructed regime, the total physical system must include ordinary matter, active relational degrees, controller/reservoir, and environment.

For every physically defined conserved quantity Q_k ,

$$\Delta Q_{k,D} + \Delta Q_{k,R} + \Delta Q_{k,\text{env}} = 0.$$

This is the governing conservation firewall.

A route has a natural state-transport map

$$Q_{\text{nat},B} = \mathcal{T}_{AB}(Q_A).$$

If the commanded arrival state differs,

$$\Delta Q = Q_{\text{desired},B} - Q_{\text{nat},B},$$

that difference must be supplied by explicit physical reservoirs.

Thus Loom translation is not free braking.

21. Free-energy cycle theorem as a kill condition

Consider a restored cycle returning the domain, controller, reservoir, environment, and relational field to their initial macroscopic states. If the cycle yields positive extractable work,

$$W_{\text{net}} > 0,$$

with no compensating state change elsewhere, the model violates its conservation structure and is rejected. Symbolically,

$$\Delta Q_{\text{total}} = 0 \implies W_{\text{net}} \leq 0$$

for a fully restored cycle, with equality only in the reversible idealization.

This is not a new physical theorem; it is the explicit thermodynamic hard-kill criterion imposed on every proposed Loom exploit.

22. Chronology attack

Unrestricted Lorentz-symmetric controllable FTL permits tachyon-antitelephone constructions. A spacelike signal with effective speed $u > c$ has, under a boost (V),

$$\Delta t' = \gamma L \left(\frac{1}{u} - \frac{V}{c^2} \right),$$

so event ordering reverses when

$$V > \frac{c^2}{u}.$$

For ($u=13c$), this threshold is only

$$V > c/13 \approx 0.0769c.$$

Reversed ordering in one frame is not alone a causal loop, but suitable two-leg constructions can close one. Therefore “no FTL radio” and “preprogram the ship” are not chronology solutions.

LOOM’s governing response is stronger: the accessible M1 manifold contains only one causal sector, and every controllable transition satisfies

$$\Delta T_R > 0.$$

A sequence of (n) controllable transitions obeys

$$T_{R,n} - T_{R,0} = \sum_{i=0}^{n-1} \Delta T_{R,i} > 0,$$

so it cannot return to an earlier relational-order state.

The unresolved theoretical burden is explaining why the accessible sector has this orientation without spending M3.

23. Destination admissibility

M1 does not imply arbitrary endpoint resolution. Let the effective uncontrolled action decompose schematically as

$$B_j^0 = b_{\text{route}} + B_{\text{geom}} + B_{\text{env}} + B_{\text{identity}} + B_{\text{boundary}} + \dots$$

This is a bookkeeping decomposition, not a proven microscopic sum.

Occupied macroscopic matter, incompatible curvature, causal trapping, excessive identity strain, or poor boundary closure can raise the action or eliminate a globally composable endpoint entirely.

This supplies the physical basis for no-telefrag behavior without adding a separate anti-collision miracle.

24. Strong-field constraint

The framework preserves ordinary causal trapping. A domain inside a region from which no future-directed outward causal completion exists is not granted a stable outward Loom solution merely by M1.

This is deliberately conservative: black-hole one-way structure is not casually erased.

A future parent theory must show how relational embeddings interact with horizons. Until then, outward escape from causally trapped domains is prohibited.

25. Observable program in 2226

A foundational theory becomes useful only when it tells experimenters what to measure.

The minimum program is:

1. **Identity channel tomography:** estimate ϵ_I for a hierarchy of protected observables.
2. **Boundary leakage:** measure ϵ_{leak} versus geometry, attachment state, temperature, and control authority.
3. **Transition kinetics:** fit $\ln \Gamma_j$ versus control bias to test exponential-action behavior.
4. **Prefactor spectroscopy:** measure Ω_j across domain classes and routes.
5. **Memory tests:** vary pulse history to delimit the Markovian regime.
6. **Route reciprocity tests:** compare $A \rightarrow B$ and $B \rightarrow A$ while separating deeper coherence from operational accessibility.
7. **Conservation calorimetry:** close energy, momentum, angular momentum, charge, radiation, and wake accounts.
8. **Metric response spectroscopy:** fit continuous embedding response and reject forced linearity when residuals demand it.
9. **Chronology null tests:** deliberately search for configurations predicted to reduce T_R ; failure must be empirical, not administrative.
10. **Blind topology inference:** infer hidden route structure from held-out transition data and score predictions prospectively.

26. Falsification and hard kills

The following outcomes kill or reopen the foundations:

- M1 reduces to gauge redundancy.
- M2 requires an independent energy or momentum exception.
- A stable ordinary-physics IR phase is structurally incompatible with the parent candidate.
- Identity preservation requires destruction/reconstruction of the protected algebra.
- Domain membership cannot be defined without arbitrary external labels.
- A restored cycle yields positive net work.
- Both causal orientations are controllably accessible and permit a closed causal loop.
- Metric transport requires an additional independent exotic-stress-energy miracle.
- Nonlocal translation requires a hidden traversed corridor.
- A successful transition produces multiple independently usable copies.
- The operating region collapses to measure-zero fine tuning once stability and realistic perturbations are included.

The project rule is not “save the theory.” It is “find the smallest theory that survives.”

27. What is derived, and what remains fictional

Established or standard mathematical machinery

- gauge quotienting;
- effective actions and Hessian stability;
- path-integral/saddle logic;
- operator algebras and quantum channels;
- error-correction/preserved-information mathematics;
- no-cloning;
- moduli-space effective dynamics as a general technique;
- semiclassical exponential transition suppression;
- competing-rate/Poisson mathematics;
- Noether/Ward conservation logic;
- GR metric/geodesic analysis;
- chronology problems associated with unrestricted FTL.

Explicit fictional content

- M1: macroscopic degenerate physical embedding.
- M2: controllable relational susceptibility.
- existence of a suitable accessible embedding moduli space for macroscopic domains.
- Mc/NRE as an implementation of M2.
- the actual hidden relational graph/topography.
- the measured constitutive coefficients of 2226 devices.

Open deep-model questions

- microscopic parent ontology;
- derivation of a 3+1 Lorentzian GR/QFT phase;
- natural emergence of bounded subsystems;
- microscopic origin of causal-sector orientation;
- rigorous instanton solution for a macroscopic Loom transition;
- full identity-channel selection/measurement theory;
- exact interaction of relational degrees with horizons;
- whether the parent-action candidate is more than a useful scaffold.

28. Discussion

The framework's strongest claim is not that it has solved quantum gravity. It has not.

Its strongest claim is methodological: **two fictional postulates are sufficient to define a constrained effective research program without granting arbitrary magic downstream.**

M1 changes what counts as physically possible. M2 changes what can be controlled. Everything else is forced to pass through identity, boundary, causality, conservation, stability, environmental compatibility, and measurement.

This structure also explains why continuous metric travel and discontinuous Loom translation need not be independent miracles. If the physical state has an embedding moduli space, smooth motion within a connected branch and nonperturbative transition between separated basins are two dynamical regimes of the same underlying degree of freedom.

That unification is attractive, but it remains vulnerable. In particular, the parent action, the emergence of the relational time orientation, and the natural definition of macroscopic domains are not yet derivations from a complete microscopic model. They are precisely the places a hostile theoretical physicist should attack.

29. Conclusion

The minimal foundational spine is

ordinary IR physics + M1 + M2

which yields, conditionally,

embedding moduli + protected identity + dynamical domains

and two operating regimes:

continuous branch motion $\rightarrow$ subluminal metric transport

nonperturbative branch transition $\rightarrow$ Loom translation.

The theory remains scientifically honest only while the boundary between established mathematics and fictional physics stays explicit. In particular, causal-sector orientation is currently a governing effective restriction whose microscopic derivation remains open; it is not counted as solved merely because the operating MVP requires it.

The appropriate claim is therefore not:

“Modern physics predicts the Loom.”

It is:

“Given two explicitly fictional extensions, the Loom can be formulated as a falsifiable effective theory whose downstream consequences remain answerable to ordinary mathematics, conservation, stability, and experiment.”

That is the standard Paper I is intended to establish.

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Appendix A — Project-source provenance

This paper synthesizes the current LOOM project foundations. The governing engineering-facing baseline freezes M1/M2, protected identity, causal-sector scope, conservation, duplication prohibition, and the distinction between Loom and the promoted subluminal metric branch. The earlier graduate hostile-review

scaffold goes deeper than the engineering MVP by proposing a pregeometric parent action, global composability, dynamical autonomy, embedding moduli, an emergent relational time orientation, and instanton-style transitions. Those deeper structures remain research scaffolds unless separately promoted.

Appendix B — Notation

Symbol	Meaning
$\$D\$$	dynamically controlled physical domain
$\mathcal{A}_D$ observab	le algebra associated with the domain
$\mathcal{A}_I$ protecte	d identity algebra
$\$E\$$	external relational embedding
$\mathcal{M}_E$ moduli s	pace of physically inequivalent embeddings
q_R	accessible relational causal-sector label
T_R	relational ordering parameter
$\mathbf{Q}$ ordinary	conserved/state variables
ϕ primitiv	e relational degree in the parent candidate
S_R	parent relational action
J_A	engineered relational source
$\mathcal{J}$ engineer	ing control authority
h^a l	ocal embedding-moduli coordinates
G_{ab} e	ffective metric on embedding moduli
B_j	dimensionless transition action to endpoint (j)
Ω_j attempt-f	requency prefactor
Γ_j transitio	n rate
ϵ_I identity-	preservation error
ϵ_{leak} dynamical domain	leakage
$\mathcal{T}_{AB}$ natural a	rival-state transport map

LOOM 2226 — Paper II

Engineering Consequences of Relational Transport

Active-medium architecture, Loom boundary control, subluminal metric transport, torch closure, and integrated courier certification

Version: 1.0

Date: 20 August 2026

Status: Engineering research paper / canon-facing synthesis

Canon compliance: Audited against Canon Baseline Manifest v2.2, CANON I v2.1.1, CANON II v2.2, CANON III v2.1.1, and their explicit governing dependency chain. Research extensions do not silently amend canon. **Foundational dependency:** Paper I — *Mathematical Foundations of Relational Embedding*

Governing project authority: Canon Baseline Manifest v2.2; Conceptual Physics Engineering MVP v1.3; Loom WP-1-WP-5; Metric WP-8-WP-11; Inherited Courier Integrated Propulsion Baseline v1.1; CANON II v2.2; CANON III v2.1.1.

Abstract

Paper I introduced a two-postulate effective framework in which a bounded physical domain can possess multiple gauge-inequivalent external embeddings while preserving a protected internal observable algebra (M1), and in which specially prepared matter can exert finite controllable authority over the relational degrees of freedom that distinguish those embeddings (M2). Paper II asks the engineering question that follows: **what must a real vehicle look like if those postulates are true but ordinary mechanics, thermodynamics, radiation transport, metrology, conservation, and failure analysis remain binding?**

The result is a three-regime propulsion architecture. Ordinary torch propulsion performs local momentum exchange, orbital mechanics, docking, acceleration, braking, rescue, and arrival-state matching. A continuous subluminal metric branch uses the same Mc-299m/nuclear-relational-exciton (NRE) order parameter to produce calibrated continuous embedding displacement through a smooth transport geometry. A discontinuous Loom branch prepares a dynamically autonomous domain and biases a finite-action transition to a certified relational endpoint. Metric and Loom share hardware and active material but are not simultaneously crew-rated.

The current reference courier closes at approximately 1,158.5 t wet mass with 250 t normal reaction mass and an 88 t unified relational plant containing 10.0 kg of Mc-299m in 100 active tiles and 208 distributed nodes. The metric branch is certified at 0.268c, 0.437c, 0.519c, and 0.595c for NORMAL through HARD modes; the Loom branch uses a normal controlled boundary of approximately 1,900 m², target transition action $B_T = 16$, 10^9 s^{-1} attempt frequency, 0.25 s commit interval, target/competitor action separation 40, and boundary leakage target 10^{-12} per gate.

The paper distinguishes closed ship-level engineering from open microphysics. In particular, coarse 2226 Mc-299m production/recovery throughput is now promoted canon, while the exact industrial synthesis pathway, producer shares, and long-term aging remain open; the full four-dimensional finite renormalized stress-energy benchmark for the metric ramp remains a standing research target; and the torch's few-parts-per-million ship-deposition requirement is system-level closed but not yet microphysically closed at the reactor/nozzle level. The central engineering conclusion is that relational transport does not make the ship less physical. It makes **configuration control, cryogenics, metrology, thermal rejection, boundary knowledge, and certification** first-class propulsion hardware.

1. Scope and epistemic standard

This paper inherits the epistemic labels of Paper I:

- **E — Established:** ordinary contemporary physics or engineering mathematics.
- **A — Analogy:** established formal machinery used as an engineering analogy.
- **P — Postulate:** M1/M2 or a direct fictional consequence of them.
- **C — Constitutive closure:** an in-universe measured/calibrated engineering law.
- **D — Derived:** follows from the stated assumptions and calibration.
- **O — Open:** unresolved and forbidden from silently carrying design load.

The purpose is not to prove relational transport from contemporary physics. The purpose is to show that, once M1 and M2 are granted, the vehicle can be engineered without quietly adding free momentum, free heat disposal, infinite control precision, arbitrary metric editing, destructive teleportation, or a third miracle.

2. Governing engineering architecture

The reference vehicle contains three distinct mobility regimes.

Regime	Physical role	Conservation mechanism
Torch	docking, orbital mechanics, acceleration, braking, close maneuver, rescue, combat	ordinary momentum exchange with expelled reaction mass
Metric	continuous sub-c long-baseline transport in characterized space	continuous M2-driven embedding response plus explicit background/field accounting
Loom	discontinuous interstellar relational re-embedding	natural arrival-state map plus explicit external exchange for mismatch

No regime silently performs another regime's job.

A Loom arrival does not grant free velocity matching. A metric field is not a docking thruster. A torch does not solve relational topology. The operating doctrine is therefore a consequence of physics, not merely gameplay differentiation.

3. Unified relational propulsion plant

The current reference courier uses a shared Mc/NRE plant rather than separate legacy Loom and metric systems.

Quantity	Reference value
Unified plant mass	~88 t
Active Mc-299m	10.0 kg
Active tiles	100 × 100 g
Distributed nodes	208
Shared reversible field bank	2 GJ
Cryoplant	150 kW-class at 20 K
High-drive thermal reject interface	900 K

The operational transition is:

METRIC COLLAPSE → NRE RELAX/REPHASE → LOOM FORMATION

There is no crew-rated direct mid-field metric-to-Loom switch.

This shared architecture is important. If metric and Loom required unrelated exotic materials, unrelated couplings, and unrelated field hardware, the setting would have spent more than its two-miracle budget. Sharing the same order parameter makes the two technologies engineering regimes of M2 rather than separate fictional discoveries.

3.1 Mc-299m strategic-material constraint

CANON II v2.2 promotes the coarse 2226 supply state for flight-grade Mc-299m:

$$\dot{M}_{\text{primary}} \approx 1,000 \text{ kg y}^{-1},$$

$$M_{\text{cumulative}} \approx 8.7 \text{ t},$$

with normal recovered/requalified return flow

$$\dot{M}_{\text{recovered}} \approx 150\text{--}250 \text{ kg y}^{-1}.$$

The working central serviceable flow is therefore approximately

$$S_{\text{Mc}} \approx 1.2 \text{ t y}^{-1}.$$

Qualified demand modestly exceeds this normal serviceable flow at the margin. The engineering consequence is not fuel starvation: Mc is reusable capital. Instead, certification queues, retrofit deferrals, strategic reserves, export controls, recovery/requalification, and pressure to reduce active inventory become part of fleet architecture. Exact plants, producer shares, industrial synthesis physics, and long-term aging remain open.

4. Mc-299m as an active medium

4.1 Scientific scaffold and fictional seam

Mc-299m is a fictional flight-grade active medium. Contemporary nuclear physics does not establish a long-lived, industrially manufacturable Mc-299m isomer with relational susceptibility.

The ordinary-science scaffold is narrower:

1. ^{299}Mc has $Z = 115$, $N = 184$, placing it at a neutron number associated with predicted superheavy shell structure.
2. Calculations of superheavy nuclei show that high- K or multi-quasiparticle configurations can substantially hinder decay in some cases [1,2].
3. Coherent collective nuclear excitations and coherent optical/X-ray control of nuclear ensembles in solids are experimentally real [3].
4. **M2 is the fictional step:** the driven collective nuclear/lattice state acquires relational back-reaction.

Thus the engineering statement is not “real nuclear physics predicts Mc drive material.” It is “real nuclear physics supplies enough structure to make the active medium a specific falsifiable material problem rather than a magic crystal.”

4.2 Active inventory

For Mc mass M_{Mc} ,

$$N_a = f_q \frac{M_{\text{Mc}}}{M_{299}} N_A.$$

Reference qualification uses

$$f_q = 0.90, \quad M_{299} = 0.299 \text{ kg mol}^{-1}.$$

Therefore one kilogram of flight-grade material contains approximately

$$N_a \approx 1.81 \times 10^{24}$$

qualified active nuclei.

The reference composite uses an effective Mc loading

$$\rho_{\text{Mc,eff}} = 100 \text{ kg m}^{-3},$$

giving

$$n_a \approx 1.81 \times 10^{26} \text{ m}^{-3}.$$

This is composite loading, not a prediction of elemental Mc density.

4.3 Nuclear-relational exciton state

The driven state is the **NRE** — **nuclear-relational exciton**.

The ordinary-physics portion is a phase-coherent collective nuclear/resonator state involving prepared sublevel populations, hyperfine and strain tuning, resonator occupation, bias fields, and phase locking. The relational back-reaction is M2.

A hard safety condition follows:

Normal M2 operation may not require bulk discharge of the Mc nuclear-isomer energy.

If useful authority depends on burning down the isomer inventory, the present Mc implementation fails. Mc is intended to be reusable active capital stock.

5. NRE constitutive response

Define

$$x = \sqrt{\frac{2W_a}{N_a \epsilon_*}},$$

and

$$\mathcal{J}_{\text{eq}} = \mathcal{J}_* N_a C_T \frac{x}{\sqrt{1+x^2}}.$$

The terms are interpreted as:

- W_a : recoverable energy in the driven collective/resonator state;
- ϵ_* : effective control-energy scale per participating nucleus;
- $\mathcal{J}_*$: M2 action-bias coupling per coherent nucleus;

- C_T : thermodynamic/order-parameter availability.

Reference first-generation calibration is:

Parameter	Value
ϵ_*	100 eV/site
T_c	60 K
T_{op}	20 K
τ_R	0.25 s
τ_{loss}	10^4 s
$\mathcal{J}_*$	10^{-22} per site

At 20 K,

$$C_T = \sqrt{1 - (20/60)^2} \approx 0.943.$$

For 1 kg Mc at $x = 1$,

$$W_a \approx 14.5 \text{ MJ},$$

$$\mathcal{J}_{\text{eq}} \approx 121,$$

and state-loss heat is approximately

$$P_{\text{state-loss}} \approx 1.45 \text{ kW}.$$

The constitutive law saturates:

$$\frac{\mathcal{J}}{\mathcal{J}_{\text{max}}} = \frac{x}{\sqrt{1 + x^2}}, \quad W_a \propto x^2.$$

Consequently, routine engineering prefers more qualified coherent material to extreme overdrive when geometry and mass permit it.

6. Array coherence and fault topology

For tile authority $\mathcal{J}_i$ and phase ϕ_i ,

$$Q_{\text{array}} = \frac{|\sum_i \mathcal{J}_i e^{i\phi_i}|}{\sum_i \mathcal{J}_i}, \quad 0 \leq Q_{\text{array}} \leq 1,$$

and

$$\mathcal{J}_{\text{array}} = Q_{\text{array}} \sum_i \mathcal{J}_i.$$

This immediately makes phase coherence a propulsion variable.

Random isolated node loss and a contiguous cluster loss are not equivalent. A spatially clustered defect can damage one wall sector, one geodesic-shaping lobe, or one Loom boundary segment before the raw percentage of failed nodes appears alarming.

Crew certification therefore tracks:

- spatial contiguity;
- local phase error;
- local authority reserve;
- affected boundary/wall sector;
- current drive mode;
- downstream geodesic, caustic, or boundary-leakage consequence.

This is a general engineering lesson: **distributed control systems fail geometrically, not merely numerically.**

7. Safe/off state

Without coherent phase preparation, microscopic contributions add approximately as a random walk:

$$\mathcal{J}_{\text{off,rms}} \sim \mathcal{J}_* \sqrt{N_a}.$$

For 1 kg reference Mc,

$$\sqrt{N_a} \approx 1.35 \times 10^{12},$$

so

$$\mathcal{J}_{\text{off,rms}} \sim 1.3 \times 10^{-10},$$

compared with commanded authority of order 121.

The on/off ratio is therefore of order

$$10^{12}.$$

Temperature provides another interlock: above T_c , the reference constitutive envelope gives $C_T \rightarrow 0$.

A ship does not accidentally Loom merely because it carries Mc. It requires cold qualified material, coherent preparation, valid phase geometry, boundary control, target solution, and commit authority.

8. Loom boundary engineering

Paper I defines the translated object as a dynamically autonomous domain rather than a narratively selected volume.

At ship level, boundary engineering must reduce leakage to the crew-rated target:

$$\epsilon_{\text{leak}} \leq 10^{-12}$$

per committed gate.

The current courier's normal controlled boundary is approximately

$$A_D \approx 1900 \text{ m}^2,$$

with an extended mature-route envelope of approximately

$$A_{\text{ext}} \approx 2430 \text{ m}^2.$$

The normal effective patch count is approximately

$$N_{\text{eff}} \approx 971,$$

with normal boundary authority approximately

$$\mathcal{J}_B \approx 670.$$

These values are calibrated architecture values, not universal constants.

8.1 Full-array authority reconciliation

The per-kilogram NRE example in Section 5 is a constitutive reference point, not the final ship ledger. For the 10 kg reference inventory at $x = 1$, the coherent full-array gross authority before reserve is approximately

$$\mathcal{J}_{\text{gross}} = 10(121)Q_{\text{array}},$$

with reference array coherence $Q_{\text{array}} \approx 0.97$, giving

$$\mathcal{J}_{\text{gross}} \approx 1.17 \times 10^3.$$

The ship does **not** assign all of this to one function. Loom formation must carry boundary authority, target/transition authority, control margin, degraded-state reserve, and calibration uncertainty. The normal boundary allocation is

$$\mathcal{J}_B \approx 670,$$

leaving the remainder for route/transition authority and reserve under the certified operating solution. Metric operation uses the same array in a different phase solution and is calibrated independently through $\mathcal{J}_M$ and χ_M ; metric and Loom authority numbers are therefore not added simultaneously.

This closes the apparent 121-per-kg versus 670-full-ship mismatch: **121 kg⁻¹** is a material-state reference; **670** is one allocated engineering burden within a coherent 10 kg array, not its gross authority. A future machine-readable engineering companion should publish the exact mode-by-mode authority partition and reserve ledger.

The engineering consequence is that large-domain difficulty is driven at least as strongly by boundary area, shape complexity, state uncertainty, node topology, cryogenic distribution, and metrology as by enclosed mass.

9. Loom transition reliability

The reference crew-rated transition uses

$$B_T = 16, \quad \Omega = 10^9 \text{ s}^{-1}, \quad \tau_g = 0.25 \text{ s}.$$

The target rate is

$$\Gamma_T = \Omega e^{-B_T}.$$

Numerically,

$$e^{-16} \approx 1.125 \times 10^{-7},$$

so

$$\Gamma_T \approx 112.5 \text{ s}^{-1}.$$

For a single dominant target channel, the probability of no transition during the 0.25 s gate is approximately

$$P_0 = e^{-\Gamma_T \tau_g} \approx e^{-28.1} \approx 6.2 \times 10^{-13}.$$

Thus the nominal target transition probability is effectively unity at engineering precision, before competing-route and boundary-error accounting.

A target/competitor action separation

$$\Delta B_{\text{select}} = 40$$

gives an individual equal-prefactor suppression

$$e^{-40} \approx 4.25 \times 10^{-18}.$$

The important result is not the exact number. It is that finite action margins can produce extreme reliability without infinite positional precision.

10. Loom conservation and arrival-state matching

A Loom transition maps the domain to the route's natural conserved arrival state:

$$Q_A \xrightarrow{\mathcal{T}_{AB}} Q_B^{\text{nat}}.$$

If the desired state differs,

$$\Delta Q = Q_B^{\text{desired}} - Q_B^{\text{nat}},$$

that mismatch must be paid by ordinary exchange.

The practical consequence is severe and useful: **the ship remains a rocket before and after Loom.**

A destination can be relationally adjacent yet operationally expensive because the arrival frame, orbital state, local traffic geometry, or terminal environment demands substantial ordinary maneuver.

This prevents Loom from erasing orbital mechanics, ports, reaction-mass logistics, or local interception.

11. Continuous metric branch

The metric branch uses the same M2 relational sector continuously rather than as a nonperturbative gate. The effective drive-active gravitational equation is represented as

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\text{ordinary}} + \mathcal{C}_{\mu\nu}^R,$$

with

$$\mathcal{C}_{\mu\nu}^R \rightarrow 0$$

in the off state.

The engineering constitutive law is

$$\beta_{\text{machine}} = \chi_M \frac{\mathcal{J}_M}{N_{\text{eff}}}.$$

Here χ_M is **measured**, not derived.

Reference production target:

$$\chi_M = 1.00, \quad 0.95 \leq \chi_M \leq 1.05, \quad \sigma_\chi \leq 0.01.$$

If the response becomes measurably nonlinear, the correct engineering response is a fitted function or table. The model is not allowed to preserve a pretty linear equation against the data.

12. Metric geometry and real-GR constraints

The reference geometry is an axial Alcubierre-family transport field with a long forward massive/null geodesic-shaping lobe and a smooth 6 m C^3 wall.

Alcubierre's original work established a famous transport geometry in general relativity [4]. Bobrick and Martire later developed a broader classification and emphasized that warp shells do not eliminate the need for physical propulsion or stress-energy accounting [5]. Santiago, Schuster, and Visser showed that generic physically reasonable warp drives in standard GR face severe energy-condition violations [6].

LOOM therefore does not claim that standard GR supplies the drive.

The division of labor is:

$$\text{GR} \rightarrow \text{geometry, geodesics, causal structure, diagnostics,}$$

$$\text{M2} \rightarrow \text{fictional constitutive access to the required relational response.}$$

This distinction is essential. Without it, the metric branch would be an unsupported claim that an Alcubierre metric can simply be built.

13. Certified metric operating card

The current courier card is:

Mode	Certified transport	Ramp	Cold load	Cryo electrical	900 K radiator equivalent
NORMAL	0.268c	10 s	20.0 kW	3.5 MW	106 m ²
FAST	0.437c	12 s	38.1 kW	6.7 MW	202 m ²
EXPEDITE	0.519c	15 s	63.5 kW	11.2 MW	336 m ²
HARD	0.595c	30 s	136.0 kW	23.9 MW	719 m ²

The array saturation response near 0.678c is not certified.

$$\boxed{\text{FTL METRIC} = \text{HARD KILL}}$$

The speed card is not a universal property of Mc-299m. It is a calibrated performance envelope of one architecture, one material lot class, one node topology, and one certification regime.

The quoted ~0.678c saturation response is likewise a **measured upper-array response**, not the asymptote of the local linear calibration equation. The linear $\beta_{\text{machine}} = \chi_M \mathcal{J}_M / N_{\text{eff}}$ law is valid only over its certified response interval. Near saturation, the response is explicitly permitted to become nonlinear and must be represented by the measured function/table. No arbitrary tanh law is assumed merely to force subluminality.

14. Metric environmental certification

A metric ship cannot treat the external environment as if it were merely a relativistic bullet field impinging on a conventional hull.

The actual problem is trajectory propagation in the commanded geometry.

Certification inputs include:

- optical/radar/lidar tracks;
- dust and plasma sampling;
- stellar photon field;
- commanded metric form;
- deflector mode;
- array health;
- phase error;
- current wall shape and ramp state.

Certification must demonstrate:

1. massive-particle trajectories do not intersect the protected worldtube above accepted loads;
2. no dangerous null caustic or conjugate point intersects the protected worldtube;
3. wave optics replaces geometric optics near caustic regimes;
4. bright-source geometry remains inside certified limits;
5. background/Killing-energy accounting closes;
6. thermal and cryogenic limits close;
7. node topology remains within degraded-state limits.

A missed real-time certification deadline produces:

$$\boxed{\text{HOLD} \rightarrow \text{DERATE/REORIENT} \rightarrow \text{COLLAPSE}}$$

rather than “continue and hope.”

15. Metric stand-off and terminal geography

Because a high-speed metric field cannot collapse instantaneously, terminal clearance remains macroscopic.

Mode	Nominal collapse travel	Default $3\times$ clearance
NORMAL	0.40 million km	1.21 million km
FAST	0.79 million km	2.36 million km
EXPEDITE	1.17 million km	3.50 million km
HARD	2.68 million km	8.03 million km

These distances are a direct consequence of speed and finite ramp time.

The governing table assumes the certified first-order **linear collapse ramp**,

$$v(t) = v_0 \left(1 - \frac{t}{\tau}\right), \quad 0 \leq t \leq \tau,$$

so nominal collapse travel is

$$D_{\text{collapse}} = \int_0^\tau v(t) dt = \frac{1}{2} v_0 \tau.$$

At HARD,

$$v_0 = 0.595c \approx 1.784 \times 10^8 \text{ m s}^{-1}, \quad \tau = 30 \text{ s},$$

therefore

$$D_{\text{collapse}} \approx \frac{1}{2} (1.784 \times 10^8)(30) \approx 2.68 \times 10^9 \text{ m} = 2.68 \text{ million km}.$$

The default $3\times$ clearance is therefore approximately 8.03 million km. The same $\frac{1}{2}v_0\tau$ relation reproduces all four rows of the certified card.

This preserves ports, approach corridors, traffic control, and torch terminal legs despite very fast interplanetary cruise.

16. Metric semiclassical-QFT screen

For a wall thickness

$$\delta = 6 \text{ m},$$

the light-crossing time is

$$\tau_{\text{wall}} \sim \frac{\delta}{c} \approx 2.0 \times 10^{-8} \text{ s}.$$

For ramp times of 10–30 s,

$$\epsilon_{\text{ad}} \sim \frac{\tau_{\text{wall}}}{\tau_{\text{ramp}}} \sim 10^{-9}.$$

This is encouraging only in a limited sense: the commanded geometry changes extremely slowly compared with the microscopic wall-crossing timescale.

It is **not** a proof that the renormalized stress-energy tensor remains finite and benign.

The full four-dimensional time-dependent finite-RSET benchmark remains a standing research and certification target. A sub-c screen has found no current horizon/divergence hard kill; that is not permission to extrapolate to arbitrary metric editing.

17. Cryogenic closure

The NRE active state operates near 20 K. The certified card distinguishes **cold load** from **electrical input to the integrated cryogenic chain**. The current architecture is calibrated at approximately

$$\kappa_{\text{cryo}} \equiv \frac{P_{\text{elec}}}{Q_{20\text{K}}} \approx 176 \frac{W_e}{W_{\text{cold}}}$$

across the certified modes:

Mode	Cold load	Cryo electrical	Implied κ_{cryo}
NORMAL	20.0 kW	3.5 MW	175 W/W
FAST	38.1 kW	6.7 MW	176 W/W
EXPEDITE	63.5 kW	11.2 MW	176 W/W
HARD	136.0 kW	23.9 MW	176 W/W

Status: C — measured/calibrated first-generation cryoplant performance.

Carnot performance supplies only a thermodynamic lower bound. For an ideal refrigerator operating between a 20 K cold stage and a hot-side temperature T_h ,

$$\text{COP}_{\text{Carnot}} = \frac{T_c}{T_h - T_c}, \quad \left(\frac{P_{\text{work}}}{Q_c} \right)_{\text{ideal}} = \frac{T_h - T_c}{T_c}.$$

For example, a 20 K to 300 K ideal stage requires at least

$$\frac{P_{\text{work}}}{Q_c} = 14,$$

while a 20 K to 900 K ideal single-stage bound would be

$$\frac{P_{\text{work}}}{Q_c} = 44.$$

Neither number **derives** the certified 176 W/W card. The larger empirical ratio includes finite real-machine efficiency, staging, pumping/conversion losses, plumbing, controls, and installed margin. The detailed internal hot-side staging is not fixed by current canon and is therefore not reverse-engineered here from the 900 K final rejection interface.

The correct engineering claim is consequently narrower and reproducible: the current 2226 cryoplant must demonstrate $\kappa_{\text{cryo}} \approx 176$ W/W over the certified load range, while Carnot sets a non-negotiable lower bound. A future detailed cryogenic work package should close the stage temperatures and fractions of Carnot explicitly.

18. Radiator closure

Radiative heat rejection follows ordinary Stefan-Boltzmann physics:

$$P = \epsilon \sigma A (T_{\text{rad}}^4 - T_{\text{sink}}^4).$$

For deep-space first-pass sizing where $T_{\text{sink}} \ll T_{\text{rad}}$,

$$A \approx \frac{P}{\epsilon \sigma T_{\text{rad}}^4}.$$

At

$$T_{\text{rad}} = 900 \text{ K}, \quad \epsilon = 0.9,$$

the radiative flux is approximately

$$q'' \approx 3.35 \times 10^4 \text{ W m}^{-2}.$$

Therefore 23.9 MW requires approximately

$$A \approx 713 \text{ m}^2,$$

consistent with the 719 m² HARD reference after rounding/design allowance.

This is an important closure because it shows the metric cryogenic burden is large but finite.

19. Ordinary torch propulsion

The torch remains conventional in its momentum accounting.

$$F = \dot{m} v_e,$$

and, for a kinetic exhaust approximation,

$$P_{\text{jet}} \approx \frac{1}{2} \dot{m} v_e^2 = \frac{1}{2} F v_e.$$

The ideal vacuum rocket equation is

$$\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right).$$

These are ordinary mechanics.

NASA fusion-propulsion studies identify exhaust velocity, available energy, and jet specific power as central system attributes and note why fusion is attractive for very-high-exhaust-velocity missions [7]. The LOOM torch pushes those attributes far beyond present engineering, but it does not change the momentum equation.

20. Reference courier mass ratio

Current reference wet mass:

$$m_0 = 1158.5 \text{ t.}$$

Normal reaction mass:

$$m_R = 250 \text{ t.}$$

Thus

$$m_f = 908.5 \text{ t.}$$

The mass-ratio logarithm is

$$\ln\left(\frac{1158.5}{908.5}\right) \approx 0.2431.$$

Therefore the ideal full-remass Δv card is:

Mode	v_e	Ideal Δv
ECON	3,000 km/s	~729 km/s
CRUISE	2,000 km/s	~486 km/s
EXPEDITE	1,000 km/s	~243 km/s
FAST	700 km/s	~170 km/s
HARD	450 km/s	~109 km/s
LIMIT	300 km/s	~73 km/s

The trade is explicit: higher exhaust velocity buys Δv but raises power demand for a given thrust.

21. Fusion fuel is not reaction mass

The courier's hundreds of tonnes of water/working fluid and reaction mass must not be confused with fusion reactants.

Fusion fuel supplies energy.

Reaction mass supplies expelled momentum. In the governing 300 t working-fluid/water ledger, the **250 t normal remass is a subset, not an additional mass line**:

- 150 t dedicated conditioned torch feed;
- 100 t convertible water;
- 50 t protected water reserve normally retained for shielding, life support, and thermal contingency.

Thus normal available remass is 250 t, while the captain may deliberately cannibalize some of the protected 50 t only as an emergency trade. This is why the wet-mass statement closes as 858.5 t dry plus 300 t working fluid/water, not $858.5 + 300 + 250$ t.

Reaction mass supplies expelled momentum.

This distinction is fundamental because high-specific-energy fusion can make the **fuel mass** small while the **momentum requirement** still demands substantial expelled mass.

Any ship ledger that merges the two hides the actual rocket physics.

22. Torch thermal deposition

The major unresolved conventional-engineering challenge is not the rocket equation. It is preventing multi-terawatt jet power from becoming ship heat.

Define

$$P_{\text{ship}} = f_{\text{dep}} P_{\text{jet}},$$

where f_{dep} is the fraction of jet/reactor power deposited into ship systems.

At a representative high-output jet power

$$P_{\text{jet}} = 13.47 \text{ TW},$$

one part per million corresponds to

$$13.47 \text{ MW}.$$

Thus:

Deposition	Ship heat at 13.47 TW
4 ppm	53.9 MW
8 ppm	107.8 MW
20 ppm	269 MW
50 ppm	674 MW
100 ppm	1.347 GW

The current doctrine treats approximately 4–8 ppm as the sustained regime, with 20/50/100 ppm increasingly transient FAST/HARD/LIMIT territory.

This is an extraordinarily demanding assumption.

23. Why the torch is not yet microphysically closed

The ship-level thermal arithmetic is reproducible. The reactor/nozzle physics is not yet sufficiently decomposed.

The required deposition fraction should be written as a sum of physical channels:

$$f_{\text{dep}} = f_{\text{plasma}} + f_n + f_{\gamma} + f_{\text{brem}} + f_{\text{synch}} + f_{\text{magnetic}} + f_{\text{conduction}} + \cdots$$

A credible closure must separately estimate:

- charged fusion-product interception;
- neutron production and capture;
- prompt gamma load;
- bremsstrahlung;
- synchrotron/cyclotron radiation;
- nozzle-field losses;
- magnet and first-wall heating;

- plasma instabilities and off-axis strike;
- conductive paths into structure;
- shielding secondary radiation.

Until those terms sum below the sustained ppm budget with margin, the correct status is:

Torch system thermal closure: closed at ship-budget level; open at reactor/nozzle microphysical level.

That is Paper II's largest ordinary-physics vulnerability.

24. Integrated mass statement

The governing courier reference is approximately:

Quantity	Value
Main body	$\sim 57 \text{ m} \times 9 \text{ m}$
Normal crew	6
Payload target	50 t
Working fluid/water	300 t
Normal reaction mass	250 t
Wet mass	$\sim 1,158.5 \text{ t}$
Dry excluding working fluid/water	$\sim 858.5 \text{ t}$
Unified relational plant	$\sim 88 \text{ t}$

The key engineering point is that relational hardware is substantial but not ship-dominating.

The design remains recognizably a spacecraft: structure, shielding, propulsion, stores, radiators, power conversion, sensors, life support, cargo, and maneuver systems still dominate much of the architecture.

25. Configuration-state engineering

A LOOM ship is not equally capable in every physical configuration.

Configuration	Dominant constraint
Docking	fine maneuver, capture geometry, torch inhibited drive alignment, plume exclusion, high-temperature rejection
Torch cruise	
Metric cruise	certified mass distribution, node topology, environment solution
Loom preparation/commit	minimized/frozen boundary state, precise metrology, target solution
Combat/cold run	radiator retraction and rapidly accumulating thermal debt
Atmospheric/landing	aero/thermal geometry incompatible with many spaceflight configurations

The statement “the equipment is attached to the ship” is not equivalent to “the equipment is inside the certified translation domain.”

CANON II v2.2 configuration-lock rule. A crew-rated commit solution is certified for a specific domain membership, attachment state, mass-state model, node topology, and hull/lattice configuration. Once the

sequence enters its committed interval, that configuration is fixed for the attempt. Severing a tether, releasing cargo, changing docking state, deploying hardware, or otherwise changing domain membership invalidates the solution and forces abort/quench/fault handling, **not** a clean momentum-transfer, braking, cargo-shedding, or domain-severance channel. Configuration can be deliberately changed only after the attempt has settled or been safely quenched and a new mass-state solution has been certified. The detailed transient failure physics remains open.

External booms, radiators, drones, tethers, cargo fixtures, or partially docked craft can alter domain closure and therefore change whether a Loom commit is legal.

26. Power, energy, and thermal debt

Power and energy are different engineering quantities:

$$E = Pt.$$

A 2 GJ field bank can provide 200 MW for 10 s, 20 MW for 100 s, or 2 MW for 1,000 s in the ideal accounting limit. It does not provide “2 GW” as a persistent capability.

Likewise, radiator retraction does not erase heat.

If unrejected heat accumulates in a thermal store with usable capacity E_{th} ,

$$t_{\text{reserve}} = \frac{E_{\text{th}}}{P_{\text{excess}}}.$$

This makes combat/cold-running a clock rather than a cinematic free mode.

The design doctrine is:

Power is capability; heat is the bill.

27. Reliability model

Relational propulsion introduces failure modes that conventional propulsion does not:

NRE/material

- isomer inventory loss;
- daughter accumulation;
- radiation damage;
- resonator detuning;
- phase slip;
- thermal excursion;
- strain/hyperfine drift.

Loom

- boundary leakage;
- incorrect domain membership;
- target overlap degradation;
- route-model error;
- insufficient action separation;
- commit during configuration change.

Metric

- local node cluster failure;
- phase-shape defect;
- environmental model staleness;
- geodesic interception;
- null-caustic formation;
- cryogenic saturation;
- ramp/collapse failure.

The safety philosophy is fail toward **loss of authority and collapse**, not toward extra uncommanded capability.

That principle should be treated as a design requirement for every controller and material subsystem.

28. Certification stack

Crew-rated relational propulsion requires a certification stack rather than a single “drive health” number.

A minimum stack includes:

1. **Material certificate:** Mc fraction, age, decay heat, spectral condition, radiation damage.
2. **Tile certificate:** NRE response, phase range, quench behavior.
3. **Array certificate:** Q_{array} , spatial authority reserve, correlated-failure map.
4. **Boundary certificate:** domain membership, leakage, structural-state uncertainty.
5. **Loom route certificate:** B_T , competitors, selectivity margin, return status.
6. **Metric geometry certificate:** wall smoothness, geodesic/null solution, bright-source geometry.
7. **Thermal certificate:** cryogenic lift, high-temperature rejection, storage reserve.
8. **Power certificate:** continuous generation, pulse bank, converter limits.
9. **Conservation certificate:** arrival-state or background-energy accounting.
10. **Operational certificate:** current configuration is one of the validated states.

This is why certification becomes infrastructure rather than paperwork.

29. Uncertainty and sensitivity

For an engineering output

$$y = f(x_1, \dots, x_n),$$

first-order uncertainty propagation gives

$$\sigma_y^2 \approx \sum_i \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2 + 2 \sum_{i < j} \frac{\partial f}{\partial x_i} \frac{\partial f}{\partial x_j} \text{Cov}(x_i, x_j).$$

For the linear metric constitutive law,

$$\beta = \chi_M \frac{\mathcal{J}_M}{N_{\text{eff}}},$$

the fractional uncertainty is approximately

$$\left(\frac{\sigma_\beta}{\beta} \right)^2 \approx \left(\frac{\sigma_\chi}{\chi_M} \right)^2 + \left(\frac{\sigma_{\mathcal{J}}}{\mathcal{J}_M} \right)^2 + \left(\frac{\sigma_N}{N_{\text{eff}}} \right)^2$$

when covariance is negligible.

The current $\sigma_\chi \leq 0.01$ requirement therefore caps one material contribution near the percent level, but array authority and effective-patch uncertainty must also be controlled.

For the torch,

$$P_{\text{ship}} = f_{\text{dep}} P_{\text{jet}}$$

makes sensitivity brutal:

$$\frac{\partial P_{\text{ship}}}{\partial f_{\text{dep}}} = P_{\text{jet}}.$$

At 13.47 TW, a change of only one ppm changes ship heat by 13.47 MW.

This is why torch deposition is a higher-priority sensitivity target than tiny refinements to the rocket equation.

30. Engineering hard kills

The following results invalidate the present architecture rather than merely requiring a software patch:

1. useful Mc authority requires bulk nuclear-isomer discharge;
 2. M2 operation catastrophically shortens Mc lifetime;
 3. warm/incoherent Mc retains dangerous N -scaling coherent authority;
 4. crew-rated boundary leakage cannot be achieved at finite authority;
 5. Loom requires free momentum reset or a hidden shipboard momentum bank;
 6. metric operation requires an independent third exotic effect beyond M2;
 7. a finite sub-c metric ramp develops unavoidable catastrophic QFT/RSET instability;
 8. required metric wall control is structurally unstable to realistic node defects;
 9. torch sustained operation cannot reach the required deposition fraction with plausible shielding/nozzle physics;
 10. a restored drive cycle produces net extractable work without compensating state change;
 11. the architecture permits controllable chronology violation;
 12. successful translation duplicates the usable protected domain.
-

31. Standing research targets

The engineering baseline is strong enough to build and operate the setting, but not academically finished.

The highest-value remaining work is:

- **Mc production physics:** synthesis chain, cross sections, yield, isomer population, separation, decay chain, half-life, passive heat, industrial throughput.
- **Mc aging:** daughter accumulation, lattice damage, resonator drift, refurbishment.
- **Torch microphysics:** explicit radiation/plasma/nozzle deposition closure to the ppm level.
- **Metric RSET:** full four-dimensional finite renormalized stress-energy calculation for the actual sub-c ramp family.
- **Metric environment:** route-specific massive/null trajectory maps and wave-optics behavior near caustics.
- **Node reliability:** manufactured spatial failure statistics and correlated-cluster models.
- **Loom scaling:** large-domain coherence, metrology, structural-state uncertainty, and certification complexity.
- **Arrival-state maps:** empirical $\mathcal{T}_{AB}$ characterization by route and endpoint.

- **Integrated uncertainty:** Monte Carlo propagation through material, array, thermal, environment, and route models.

32. Discussion

The engineering consequence of relational transport is almost the opposite of effortless science-fiction travel. The drive does not make ordinary engineering disappear. It creates additional things that must be engineered precisely.

The vehicle must know what belongs to it. It must know its own boundary. It must know the state of hundreds of distributed nodes. It must maintain a 20 K coherent active medium while rejecting megawatts at hundreds of kelvin. It must solve the external environment through a nontrivial geometry. It must know which relational route is actually certified. It must preserve a protected internal state. It must account for arrival momentum and energy. It must keep a multi-terawatt torch from depositing more than a few parts per million of its power into the ship.

This produces a useful hierarchy of technological difficulty:

$$\boxed{\text{physics miracle} < \text{materials} < \text{metrology/control} < \text{thermal/reliability} < \text{certification at scale}}$$

once M1/M2 are granted.

The rare Mc inventory matters strategically, but it is not consumed like propellant. Boundary precision, cryogenic plant, distributed control, route knowledge, and verified system state become the recurring operational cost.

The result is a civilization in which a ship can cross relational interstellar adjacency while still being defeated by a bad radiator, a contaminated material lot, a stale route model, a cluster of failed nodes, insufficient remass, or a terminal orbit it cannot afford to match.

That is not an incidental aesthetic choice. It is the direct engineering consequence of refusing to let the miracle do work that ordinary physics should still demand.

33. Conclusion

Given M1 and M2, a finite engineering architecture exists without adding a third miracle.

The first-generation chain is

$${}^{299m}\text{Mc}/\text{NRE} \rightarrow \mathcal{J}_{\text{array}} \rightarrow \begin{cases} \text{continuous metric authority,} \\ \text{Loom boundary + transition authority.} \end{cases}$$

The continuous branch yields calibrated subluminal metric transport; the discontinuous branch yields relational translation. Both remain bounded by material condition, phase coherence, node topology, cryogenic lift, thermal rejection, environmental certification, conservation, and finite control authority.

Ordinary propulsion remains indispensable:

$$\boxed{\text{Loom ship} = \text{real rocket} + \text{relational plant}}$$

rather than a teleportation device wearing a rocket-shaped costume.

The current courier closes as a coherent engineering object at ship level. The largest remaining technical attacks are not vague. They are specific and calculable: Mc-299m production/aging, full metric RSET benchmarking, and torch reactor/nozzle deposition physics.

Those are exactly the kinds of unresolved problems a serious technical paper should expose.

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Appendix A — Reference courier engineering card

Quantity	Value
Wet mass	~1,158.5 t
Dry excl. 300 t working fluid/water	~858.5 t
Normal remass	250 t
Payload target	50 t
Unified relational plant	~88 t
Mc-299m	10.0 kg
Active tiles	100 × 100 g
Nodes	208
Shared bank	2 GJ
Cryoplant	150 kW-class, 20 K
Loom boundary	~1,900 m ² normal
Loom extended envelope	~2,430 m ²
Loom effective patches	~971
Loom boundary authority	~670
Loom target action	16
Loom attempt frequency	10 ⁹ s ⁻¹
Loom gate	0.25 s
Loom selectivity separation	40
Boundary leakage target	10 ⁻¹²
Metric NORMAL	0.268c
Metric FAST	0.437c
Metric EXPEDITE	0.519c
Metric HARD	0.595c
Metric saturation response	~0.678c, not certified
Metric wall	6 m, C^3

Quantity	Value
FTL metric	BLACK / hard kill

Appendix B — Provenance and authority notes

This paper follows the corrected v2.2 engineering lineage. The earlier legacy architecture with 32 kg Mc, 64×0.5 kg cells, a 90 t Loom package, a separate 60 t metric package, PJ-scale banks, area-times-velocity-squared scaling, and the old 0.03/0.04/0.05c metric card is non-governing.

The promoted Loom lineage is WP-1 through WP-5. The promoted metric lineage is WP-8 through WP-11. Where this paper exposes open microphysics, it does not demote the engineering calibration; it marks the boundary between a usable in-universe engineering model and the deeper conventional-physics work still required for academic closure.

Appendix C — Standing reproducibility package for journal-facing release

A journal-facing release should be accompanied by machine-readable tables or code sufficient to reproduce:

- Mc active-site and NRE authority calculations;
- Loom transition probabilities;
- boundary authority and leakage calculations;
- metric mode constitutive fits and uncertainty;
- collapse/clearance distances;
- cryogenic electrical loads;
- Stefan-Boltzmann radiator areas;
- courier rocket-equation card;
- torch deposition sensitivity;
- integrated mass/power/thermal statements;
- Monte Carlo sensitivity and correlated node-failure cases.

No exact-looking table should be treated as fully reproducible unless its numbers can be regenerated from declared inputs.

LOOM 2226 — Paper III

Economic Geography and Civilizational Dynamics of Relational Transport

Transport cost, network topology, industrial concentration, information latency, demography, institutions, and the first interstellar frontier

Version: 1.0

Date: 20 August 2026

Status: Economic/civilizational research paper / canon-facing synthesis

Canon compliance: Audited against Canon Baseline Manifest v2.2, CANON I v2.1.1, CANON II v2.2, CANON III v2.1.1, and their explicit governing dependency chain. Research extensions do not silently amend canon. **Foundational dependencies:** Paper I — *Mathematical Foundations of Relational Embedding*; Paper II — *Engineering Consequences of Relational Transport*

Governing project authority: Canon Baseline Manifest v2.2; CANON I v2.1.1; Earth & Solar System Canon Atlas v3.1; CANON II v2.2; CANON III v2.1.1. Earlier red-team/model values are provenance only where superseded or explicitly non-canon.

Abstract

LOOM 2226 is built around a causal chain: physics generates engineering; engineering generates economics; economics generates institutions; institutions generate human stories. Paper I formalized the fictional physical seam. Paper II translated that seam into finite ships, materials, thermal systems, metrology, and certification. Paper III propagates those constraints into economic geography and civilizational structure.

The central result is that relational transport does not abolish geography. It creates a **second geography** layered over metric space. Ordinary distance, orbital mechanics, terminal stand-off, reaction mass, habitat capital, resource location, and local travel remain real. Loom connectivity adds a sparse, uncertain, empirically discovered graph whose edge costs depend on route quality, certification, vessel state, risk, information age, and operational access rather than light-years alone. The nearest star is therefore not necessarily the nearest economic neighbor.

Within Sol, mature fusion, automation, and advanced fabrication lower the real cost of ordinary energy and bulk matter while shifting scarcity toward high-performance materials, location, ownership, certification, metrology, longevity, legal authority, and trusted access. The conserved 2226 census contains exactly 8,571,824,600 biological humans and 214,068,429 recognized synthetic persons; approximately 259.3 million biological humans live off Earth and are already part of the conserved biological total. Paper III uses ~66.8 TW of continuous civilization electricity as a working macroeconomic reconstruction, **not** as a promoted Atlas constant. Earlier 8.669B / ~213M and 42-65 TW / 46.5 TW figures are superseded derivation history.

Loom arrives only in the 2220s. It therefore cannot explain Solar-System demography or mature industrial geography. Instead it strikes an already old, automated, legally dense, institutionally persistent civilization. Because independent FTL communication does not exist, ships carrying information become part of the communications network. Markets, contracts, command authority, scientific knowledge, legal state, and personal relationships acquire causal age. This produces a packet civilization in which synchronization is physically costly and local delegated authority becomes structurally necessary.

The early interstellar economy is consequently dominated not by bulk colonial trade but by payloads whose value is weakly coupled to mass: topology data, certified models, software, legal instruments, intelligence, biological and material samples, specialists, scarce components, and urgent authority. The first interstellar constitutional crisis follows naturally: institutions claim jurisdiction over people and assets they cannot continuously observe, protect, synchronize, or enforce.

1. Epistemic standard

Paper III distinguishes:

- **E — Established:** accepted economic, demographic, network, or institutional mathematics.
- **A — Analogy:** established theory used as disciplined analogy.
- **P — Postulate:** inherited M1/M2 fictional premise when referenced directly.
- **C — Canon/calibration:** explicit LOOM setting value or model calibration.
- **D — Derived:** follows from stated assumptions.
- **O — Open:** unresolved and forbidden from being silently promoted.

The paper does not claim that economics uniquely predicts the 2226 setting. It asks whether the setting’s economic and institutional structure can be generated from explicit physical and engineering constraints rather than aesthetic preference.

2. The causal spine

The governing propagation is

physics → engineering → generalized transport cost → industrial geography → institutions → human incentives

The highest-leverage upstream constraints are:

1. torch propulsion moves bulk mass conventionally;
2. metric transport is continuous, subluminal, strategic, and requires large terminal stand-off;
3. Loom translation is rare, route-dependent, and domain/certification limited;
4. Mc-299m is reusable strategic capital rather than routine propellant;
5. no independent FTL communications exist;
6. route topology must be physically discovered and updated;
7. the interstellar frontier is only a few years old in 2226.

These constraints rule out a generic frictionless “post-scarcity galactic economy.”

3. Generalized transport cost

For a movement from i to j , define generalized cost

$$C_{ij} = p_{\text{direct}} + v_t T_{ij} + p_{\text{risk}} + p_{\text{cert}} + p_{\text{inventory}} + p_{\text{unreliability}} + p_{\text{legal}} + p_{\text{info}}.$$

Here:

- p_{direct} : fuel, remass, maintenance, fees, depreciation, crew, port charges;
- $v_t T_{ij}$: value of time;
- p_{risk} : expected loss/insurance burden;
- p_{cert} : inspection, metrology, route, material, and operational certification;
- $p_{\text{inventory}}$: working capital tied up during transport and uncertainty;
- $p_{\text{unreliability}}$: variance in completion/rescue/route access;
- p_{legal} : jurisdictional and compliance burden;
- p_{info} : cost imposed by stale market/legal/technical state.

Certification cost is neither a flat fee nor assumed linear in translated mass. A first-order decomposition is

$$p_{\text{cert}} = p_0 + p_R(R) + p_B(A_D, N_{\text{eff}}, g_D) + p_N(\mathcal{T}_N) + p_S(\Sigma_S) + p_E(E_{\text{env}}) + p_L(L_{\text{prov}}),$$

where p_0 is fixed attempt overhead; R route maturity; A_D, N_{eff}, g_D boundary size/patch count/geometry; $\mathcal{T}_N$ node topology; Σ_S structural-state uncertainty; E_{env} environmental certification burden; and L_{prov}

liability/provenance burden. Mass matters indirectly where it worsens structural state, boundary geometry, or arrival-state work, but tonnes enclosed are not the sole certification variable.

Headline cruise speed affects only part of T_{ij} .

A fast metric vessel may still face million-kilometre stand-off, local torch legs, queueing, inspection, and terminal synchronization. A Loom edge may have negligible metric transit time yet high certification and uncertainty cost.

Thus

$$\frac{\partial C_{ij}}{\partial v_{\text{cruise}}} < 0$$

does not imply

$$C_{ij} \rightarrow 0 \quad \text{as} \quad v_{\text{cruise}} \rightarrow \text{large.}$$

Geography survives speed.

3.1 Canon-anchored engineering cost case

A useful worked case can be built from Paper II's certified metric card **without inventing 2226 prices**. Consider two terminals separated by approximately one astronomical unit,

$$D \approx 149.60 \text{ million km.}$$

Paper II gives default terminal clearances of 1.21 million km for NORMAL and 8.03 million km for HARD. If both endpoints require those clearances, the distance available to the characterized metric cruise segment is approximately

$$D_{\text{cruise,N}} \approx 149.60 - 2(1.21) = 147.18 \text{ million km,}$$

$$D_{\text{cruise,H}} \approx 149.60 - 2(8.03) = 133.54 \text{ million km.}$$

Using the certified transport speeds,

$$v_N = 0.268c, \quad v_H = 0.595c,$$

the idealized characterized-space cruise times are

$$T_{\text{cruise,N}} \approx 30.5 \text{ min,}$$

$$T_{\text{cruise,H}} \approx 12.5 \text{ min.}$$

Thus HARD cuts the characterized-space cruise time by a factor of roughly 2.4, while its two-end default-clearance burden occupies about 10.7% of the one-AU geometry versus about 1.6% for NORMAL. Those clearance regions still require ordinary terminal access, traffic handling, field formation/collapse discipline, and environment certification.

The generalized-cost implication is therefore not that HARD is always cheaper. It is that the time-value term

$$v_t T_{ij}$$

falls substantially while several fixed or mode-sensitive terms - especially certification, risk, thermal posture, traffic access, and conventional terminal motion - do not vanish and may rise. A high-value time-sensitive movement can therefore rationally choose HARD while low-value bulk cargo may remain on slower metric or torch service.

This example is **canon-anchored in geometry and certified speed but deliberately price-free**. Monetary values, insurance premia, queue costs, and certification charges remain model variables rather than back-door canon.

4. Two overlapping geographies

LOOM civilization occupies at least two transport spaces.

4.1 Metric geography

Metric geography includes:

- physical astronomical separation;
- gravitational wells;
- orbital state;
- terminal stand-off;
- local traffic;
- resource location;
- habitat and port infrastructure;
- reaction-mass supply;
- radiation and environmental hazards.

4.2 Relational geography

Let the discovered Loom network be

$$G = (V, E),$$

where vertices are characterized endpoint domains and edges are operationally meaningful route solutions.

Assign each edge e a generalized cost c_e . Effective Loom-economic distance is

$$d_G(i, j) = \min_{p: i \rightarrow j} \sum_{e \in p} c_e.$$

Metric light-year distance does not appear as the sole argument.

Therefore it is possible that

$$d_{\text{metric}}(A, B) < d_{\text{metric}}(A, C)$$

while

$$d_G(A, B) > d_G(A, C).$$

This is the economic expression of relational topography.

5. Route discovery as capital formation

The Loom graph is not globally known.

Humanity maps it by walking it:

1. remote theory identifies candidate relationships;
2. uncrewed probes attempt first crossings;
3. destination-side measurement constrains local route families;
4. return demonstration closes the empirical loop;
5. routes age as measurements, fixtures, and endpoint conditions change.

Route knowledge is therefore a capital asset.

Let route-information stock be K_R . A simple depreciation model is

$$\dot{K}_R = I_R - \delta_R K_R,$$

where I_R is new survey/metrology investment and δ_R represents obsolescence from topology drift, endpoint change, calibration aging, and information latency.

A route map can be valuable even when nobody owns the underlying physics, because what is scarce is **validated knowledge of how to use it now**.

6. Network centrality and chokepoints

For weighted graph G , a node's economic importance need not track population.

Useful diagnostics include weighted betweenness centrality,

$$BC(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}},$$

and edge redundancy.

A small node can become strategically important if many low-cost paths traverse it, if it provides unique certification, or if it is one of few safe staging environments.

This supplies a quantitative route to chokepoint politics.

A route or node loss changes not only travel distance but:

- packet latency;
- rescue time;
- insurance pricing;
- inventory requirements;
- market synchronization;
- military response;
- scientific replication;
- legal service;
- frontier autonomy.

The hidden graph therefore has constitutional consequences.

7. Agglomeration and industrial concentration

Krugman’s core-periphery framework shows how transport costs, increasing returns, and market size can generate endogenous concentration [1].

LOOM’s corresponding mechanism is

$$\boxed{\text{scale economies} + \text{fixed certification costs} + \text{specialized knowledge} + \text{transport friction} \rightarrow \text{agglomeration}}$$

even when raw energy and common atoms are cheap.

The current strategic-material geography places highest-end Mc, lattice/metric fabrication, superconductors, semiconductors, metrology, and certification disproportionately in Earth–Moon space and selected Mars/Mercury orbital complexes. Mars maintains a broad independent industrial stack; Ceres/Belt specialize in low-gravity water/materials, reaction mass, and shipyard functions.

These locations remain partly calibration. Paper III does not claim Krugman’s model predicts Mercury. It supplies the mechanism that any claimed concentration must satisfy.

8. Input-output propagation

Let A be an input-output coefficient matrix, x gross output, and d final demand:

$$x = Ax + d.$$

Then

$$x = (I - A)^{-1}d.$$

The Leontief inverse

$$L = (I - A)^{-1}$$

measures direct and indirect production dependence.

This is the correct framework for asking whether a strategic material actually matters.

A reduction in Mc production does not directly reduce all output. It first constrains sectors that require flight-grade relational hardware, then propagates through commissioning, urgent transport, survey, defense, insurance, and frontier investment according to the relevant entries of L .

Acemoglu et al. show why asymmetric supplier networks can transmit microeconomic shocks into aggregate volatility [2]. LOOM therefore treats supplier centrality and substitutability as more important than dramatic adjectives such as “rare.”

9. Mc-299m as strategic capital

CANON II v2.2 promotes the coarse 2226 flight-grade Mc supply state:

$$\dot{M}_{\text{primary}} \approx 1,000 \text{ kg y}^{-1},$$

$$\dot{M}_{\text{recovered}} \approx 150\text{--}250 \text{ kg y}^{-1},$$

$$S_{\text{Mc}} \approx 1.2 \text{ t y}^{-1}$$

for the working central primary-plus-requalified serviceable flow, with cumulative historical flight-grade production

$$M_{\text{Mc,cum}} \approx 8.7 \text{ t}.$$

Qualified demand modestly exceeds normal serviceable supply **at the margin**. The setting therefore has a tight strategic-capital market, not an Mc fuel famine. Existing fleets remain sustainable, while new construction, retrofit, frontier expansion, research programs, and strategic-stock replenishment compete for certified material. Shortage appears through queues, premiums, sovereign reservation/export controls, recovery markets, and deferred commissioning.

Because Mc is reusable active capital, its economic role resembles scarce high-end machine capacity more than oil. For ship class s with active inventory m_s , primary production alone gives the theoretical new-inventory ceiling

$$N_{s,\text{max}}^{\text{primary}} = \frac{\dot{M}_{\text{primary}}}{m_s}.$$

For the 10 kg reference courier,

$$N_{\text{max}}^{\text{primary}} \approx 100 \text{ courier inventories y}^{-1}$$

if every kilogram went to identical new couriers, which it does not. The $\sim 1.2 \text{ t/y}$ serviceable flow corresponds to roughly 120 courier inventories per year only as a **gross material-equivalent throughput**; recovered material is heavily relevant to sustainment and all allocations compete.

Production and highest-end relational-material fabrication are broadly concentrated in Earth-Moon space and selected Mars/Mercury orbital high-end fabrication complexes. Exact plants, producer shares, year-specific jurisdictional allocations, fleet counts, industrial synthesis physics, and long-term material aging remain open.

10. Strategic-material chain

The high-margin chain is

mine $\rightarrow$ isotope separation $\rightarrow$ ultra-purification $\rightarrow$ solid-state fabrication $\rightarrow$ metrology $\rightarrow$ certification

The economic value migrates downstream as ordinary extraction becomes increasingly automated.

This generates a crucial distinction:

Bulk atoms become cheap before high-performance matter becomes cheap.

A tonne of generic structural feedstock and a kilogram of flight-certified active material can share chemical ancestry while occupying radically different economic categories.

Certification itself is not assumed permanently expensive by fiat. AI and robotics may automate large portions of inspection and proof. What can remain scarce are:

- trusted reference standards;

- liability-bearing institutions;
- rare destructive test capacity;
- access to validated fixtures;
- adversarial certification;
- legal recognition across jurisdictions;
- provenance chains;
- performance under extreme conditions.

The unresolved question is how much of the certification rent survives mature automation.

11. Energy abundance without post-scarcity

A Paper III macroeconomic reconstruction uses a working central continuous-electricity scenario of

$$P_{\text{elec,model}} \approx 66.8 \text{ TW}.$$

Status: C/O - research-model calibration, not promoted canon. The argument below depends on energy being comparatively cheap and abundant relative to 2026, not on 66.8 TW being the uniquely correct civilization total.

The earlier assumption of universal stationary D-³He fusion fails under plausible lunar supply. The red-team review found that grid-scale lunar He-3 would require regolith processing on the order of hundreds of billions of tonnes per year. Stationary generation must therefore be diversified rather than quietly leaning on a universal lunar He-3 economy.

The economic conclusion survives:

$$\text{cheap energy} \neq \text{no scarcity}.$$

Energy abundance lowers many transformation costs but does not eliminate:

- fixed location;
 - time;
 - legal authority;
 - trusted identity;
 - high-performance material quality;
 - capital ownership;
 - habitat volume;
 - heat rejection;
 - environmental capacity;
 - exclusive access;
 - social status;
 - longevity treatment;
 - risk.
-

12. Relative prices

For sector i , let unit cost be decomposed into factor shares

$$c_i = w_i c_L + e_i c_E + m_i c_M + k_i c_K + r_i c_R,$$

where the terms represent labor, energy, ordinary material, capital/finance, and scarce rent/location/certification components.

Automation and cheap energy can drive

$$c_L \downarrow, \quad c_E \downarrow, \quad c_M \downarrow$$

without forcing

$$c_R \downarrow.$$

This explains why ordinary manufactured goods may become dramatically cheaper while prime housing, orbital slots, route access, elite rejuvenation, trusted certification, and scarce institutional authority remain expensive or become more expensive.

The working household basket around 28–35% of 2026 real cost is a rough model, not locked canon. The red-team review specifically found that its housing treatment was internally inconsistent and must be rebuilt with structure cost separated from land/location rent.

13. Automation and production

Current working calibration places machine-executable 2026-equivalent task-hours around 82–90%, with a central working value near 86%. This has not yet been derived from a complete task-weighted production model.

The useful qualitative result is stronger than the exact percentage:

- routine digital work is heavily automated;
- hazardous and repetitive physical work is heavily robotic;
- industrial output decouples substantially from resident population;
- humans and synthetic persons remain important where judgment, authority, relationship, legitimacy, anomaly handling, and mission ownership matter.

This is why the Belt and outer system can support substantial industrial throughput without requiring billions of resident workers.

14. Labor, ownership, and distribution

As machine capital substitutes for routine labor, the key political variable shifts from “who has a job?” toward “who owns productive capital and who has enforceable claims on output?”

Let aggregate output be represented schematically by

$$Y = F(K_M, K_R, L_H, L_S),$$

where:

- K_M : machine/physical capital;
- K_R : knowledge, software, certified process, and relational-route capital;
- L_H : biological-human labor;
- L_S : synthetic-person labor.

Synthetic persons are persons, not owned automation. Their compensation belongs on the labor/person side of the ledger, not automatically under capital return.

The current 20–24 hour mature-Earth paid workweek is a scenario output. The red-team review found that it is algebraically consistent with the working model’s upstream assumptions but not independently validated. Likewise, labor share remains provisional and requires an explicit production/ownership/tax model.

The setting therefore does not require one universal UBI solution. Plausible equilibria include combinations of:

- social dividends;
- sovereign/public capital funds;
- pension ownership;
- cooperative capital;
- broad household equity;
- guaranteed services;
- negative income taxation;
- inheritance reform;
- licensing/resource rents;
- conventional wages;
- local political bargains.

Distribution is institutional, not automatic.

15. Demographic control

The frozen Solar-System census is a conserving ledger.

$$N_{\text{bio}} = 8,571,824,600, \quad N_{\text{synth}} = 214,068,429.$$

Approximately 259.3 million biological humans live off Earth; Earth retains about 97.0% of biological humanity. Synthetic persons are persons rather than owned automation. Exact fertility, median-age and extreme-longevity-tail parameters remain model-sensitive unless separately promoted.

16. Off-Earth biological population

The promoted ledger contains

$$N_{\text{bio,off-Earth}} \approx 259.3 \text{ million.}$$

This is already included in the conserved biological total. Representative system values are Earth orbit 64.74M, Moon 14.96M, Mars 60.29M, Mercury 9.24M, Venus 26.28M, Ceres 7.73M, Belt majors 2.26M, Jupiter 24.18M, Saturn 31.27M, Uranus 8.08M, Neptune 6.84M, Kuiper 3.22M, Oort 242, and ~206k heliocentric biological residents.

The open demographic work is no longer basic population conservation. It is cohort detail, fertility/longevity sensitivity, habitat-capital refinement, synthetic-person individuation, and future migration under the corrected transport stack.

17. Habitat construction as the off-Earth population constraint

For region r , define habitable capacity

$$H_r(t)$$

and permanent resident population

$$P_r(t) \leq H_r(t).$$

Capacity evolves as

$$\dot{H}_r = I_{H,r} - \delta_{H,r} H_r,$$

where I_H is commissioned habitat capacity and δ_H retirement/loss.

Migration demand cannot produce residents faster than pressure volume, shielding, life support, power, transport, and civic infrastructure can be built.

Therefore

$$\dot{P}_r \leq \dot{H}_r$$

whenever habitat capacity is binding.

This is why off-Earth demography must be generated by capital formation and migration, not chosen as a flavor number.

18. Longevity and institutional persistence

Long healthy lives change political economy even when population remains roughly stable.

If expected working/asset-holding life increases, then absent countervailing institutions:

- wealth compounds for longer;
- inheritance arrives later;
- elite networks persist;
- professional reputation has longer memory;
- institutional turnover slows;
- political incumbency can deepen;
- intergenerational conflict changes character.

A simple wealth accumulation identity is

$$W_{t+1} = (1 + r_t)W_t + y_t - c_t - \tau_t.$$

Increasing the number of periods over which this recursion operates magnifies the importance of returns, taxation, consumption, gifting, and inheritance law.

Thus advanced medicine is also a constitutional technology.

19. Solar-System geography in 2226

The broad economic structure is:

- **Earth:** population, finance, political legitimacy, diversified high-end services and fabrication.
- **Earth orbit/Luna:** dense industrial and metrology complex, shipyards, microgravity industry, finance, transport.
- **Mars:** largest independent off-Earth planetary civilization with broad industrial autonomy.
- **Mercury:** small but strategically heavy inner-solar specialist economy.
- **Venus:** substantial cloud-habitat civilization with atmospheric chemistry and biotech advantages.
- **Ceres/Belt:** water, reaction mass, shipyards, logistics, insurance, distributed extraction.

- **Jupiter/Saturn:** mature outer regional economies anchored by major moons and logistics networks.
- **Uranus/Neptune:** thinner but established regional economies.
- **Kuiper/Oort:** sparse deep-system networks; Oort remains expeditionary rather than a settled economy.

This geography was largely built during the torch era. Metric arrives too late to create it and Loom arrives far too late.

Technology therefore modifies inherited geography rather than replacing it.

20. Metric transport and economic compression

The corrected courier metric envelope of roughly 0.27–0.60c materially compresses long-baseline Solar travel.

However high-speed operation requires:

- surveyed clean corridors;
- large terminal stand-off;
- caustic/geodesic certification;
- healthy distributed nodes;
- cryogenic and thermal posture;
- conventional terminal maneuver.

Therefore metric primarily changes the economics of **urgency**.

It disproportionately benefits:

- senior personnel;
- emergency response;
- high-value components;
- finance;
- intelligence;
- scientific samples;
- command;
- legal process;
- time-sensitive maintenance;
- perishable information.

Bulk mass continues to care deeply about torch economics.

This yields the compact doctrine:

Torch moves mass. Metric moves urgency. Loom moves the frontier.

21. No independent FTL communications

This is one of the highest-leverage constraints in the setting.

A ship carrying data is part of the communications network.

Let information state I at node i have causal age

$$a_I = t_{\text{local}} - t_{\text{source}}.$$

Economic and institutional decisions must therefore condition not only on content but on age and provenance.

A local actor's effective state estimate can be represented as

$$\hat{X}_i(t) = \mathcal{F}(I_1, a_1, \pi_1; I_2, a_2, \pi_2; \dots),$$

where π denotes provenance/confidence.

There is no globally synchronized “now” available to institutions across the Loom frontier.

22. Packet civilization

Information moves in discrete physical opportunities.

For route $i \rightarrow j$, expected information delay is approximately

$$T_{\text{info}} = T_{\text{wait}} + T_{\text{carriage}} + T_{\text{terminal}} + T_{\text{processing}}.$$

Even if carriage through a Loom event is effectively discontinuous, waiting for a suitable vessel, route certification, departure, and local delivery can dominate.

Packet frequency therefore becomes economically meaningful.

If departures are Poisson with rate λ , expected waiting time is

$$E[T_{\text{wait}}] = \frac{1}{\lambda}.$$

A route with one reliable packet ship per day has radically different institutional properties from one with one packet per month, even if both use the same Loom physics.

23. Information arbitrage

Suppose an asset’s true state changes at source time t_0 , but market j will not learn until $t_0 + \Delta t$.

A physical carrier able to deliver authenticated information earlier has economic value related to:

$$V_{\text{info}} \sim \Delta t \times \text{state sensitivity} \times \text{capital exposed} \times \text{credibility}.$$

This is not free arbitrage forever. Markets price expected packet schedules, counterparties demand provenance, and actors can hedge.

But causal asymmetry creates persistent value for:

- authenticated market state;
- legal authority;
- model updates;
- intelligence;
- disaster reports;
- scientific results;
- contract acceptance;
- governance instructions.

The commodity is often not the data itself. It is **being the first trusted causal carrier of the data**.

24. Contracts under causal delay

A contract cannot assume continuously synchronized parties.

Useful state variables include:

- signing causal time;
- jurisdictional version;
- latest received amendment;
- packet provenance;
- acceptance deadline;
- revocation status as last known locally;
- force-majeure route conditions;
- escrow state;
- insurer synchronization state.

A contract may be valid locally while a revocation is already physically en route.

This is not a software bug. It is a consequence of finite information propagation.

Law therefore develops explicit rules for causal notice, packet priority, good-faith reliance on stale state, and competing authoritative versions.

25. Insurance as a topology sensor

Insurance pricing aggregates information about:

- route reliability;
- rescue access;
- vessel certification;
- node health;
- operator history;
- packet frequency;
- political stability;
- legal enforceability;
- cargo value;
- information age.

A simplified premium is

$$\Pi = E[L] + \rho \text{Var}(L) + c_{\text{capital}} + c_{\text{admin}},$$

where L is loss.

Because route quality changes $E[L]$ and variance, insurance markets become indirect sensors of the hidden topology.

A sudden premium increase can reveal that underwriters know something about a route before the public technical explanation arrives.

26. Early interstellar trade composition

The extrasolar frontier is only a few years old.

Bulk interstellar trade is therefore not the default.

The early comparative advantage of Loom-capable couriers lies in payloads with high value-to-mass ratio:

- topology and survey data;
- scientific samples;
- biological material;
- certified Mc/metric components;
- software/model updates;
- cryptographic/provenance material;
- legal instruments;
- intelligence;
- specialists;
- emergency medicine;
- unique cultural artifacts;
- urgent spare parts.

A rough shipment decision compares value density v_m with generalized carriage cost per unit mass c_m :

$$v_m \gg c_m$$

for the most attractive early cargoes.

Ordinary bulk commodities generally fail this test unless they are unavailable locally and strategically essential.

27. Why the small Loom courier exists

The reference courier is unusually small because early Loom economics rewards low domain burden and high-value carriage.

Pure information carriage does **not** require a biological or even crewed courier. Where route certification permits it, autonomous packet craft can economically outperform crewed ships for data-only service because they avoid life-support and personnel-risk costs. Crewed couriers retain a separate niche when the payload includes people, discretionary legal/command authority, secure physical custody, specialist judgment, adaptive survey/recovery capability, or unique samples whose handling cannot be reduced to unattended packet carriage. No artificial “human witness” rule is introduced to protect the hero-ship niche.

Its economic function is closer to a combination of:

- packet ship;
- diplomatic pouch;
- survey vessel;
- secure data center;
- specialist transport;
- high-value logistics aircraft;
- intelligence platform;
- emergency courier.

than to a bulk freighter.

This follows from both economics and engineering. Paper II indicates that Loom burden follows boundary area/geometry, effective patch count, node topology, structural-state uncertainty, metrology, and certification rather than enclosed mass alone. Paper I adds a selectivity reason. If a more complex domain presents N_c serious competing cuts or endpoint solutions, the useful action-margin scale is

$$\Delta B_{\text{select}} \gtrsim \ln \left(\frac{N_c}{p_{\text{mis}}} \right).$$

The project does **not** yet canonize a function $N_c(A_D)$ or a universal maximum Loom hull size, so this is not a quantitative size law. But increasing boundary/configuration complexity can raise both the metrology burden

and the selectivity burden. Compact vessels are therefore favored twice: they fit the early high-value/low-mass cargo market and tend to present a simpler certified relational domain. This remains a tendency, not a universal prohibition on large Loom ships.

28. Corporate concentration

Megacorporate power is generated by vertical control across multiple scarce layers:

resource → processing → fabrication → certification → transport → ports → finance/insurance.

A firm controlling only ore faces automation and commodity competition.

A firm controlling several adjacent high-fixed-cost layers can create durable market power through:

- vertical foreclosure;
- proprietary standards;
- data advantages;
- certification lock-in;
- network effects;
- insurance relationships;
- political bargaining;
- route knowledge;
- installed-base maintenance.

This is the mechanism behind megacorporate influence. It need not imply that corporations replace states.

29. State power and strategic industry

States remain powerful because many scarce goods are jurisdictional:

- legal personhood;
- property recognition;
- coercive authority;
- military force;
- taxation;
- border/access control;
- certification recognition;
- spectrum/traffic allocation;
- rescue obligations;
- sanctions;
- criminal law;
- treaty recognition.

Corporations can own ships and fabs. They cannot automatically manufacture legitimate sovereignty.

The 2226 political economy is therefore plural rather than cyberpunk-feudal by default: states, cities, habitat governments, route authorities, corporations, insurers, unions, universities, military institutions, religious bodies, and frontier communities overlap.

30. The interstellar constitutional problem

Sol institutions claim continuing authority over citizens, settlements, subsidiaries, contracts, and assets that may be causally remote.

Frontier communities ask a structurally difficult question:

What is the legitimate scope of authority that cannot continuously observe, protect, synchronize, or enforce?

Let institutional service latency be T_S and enforcement latency T_E . As these rise relative to the timescale of local decisions T_D ,

$$\frac{T_S}{T_D} \uparrow, \quad \frac{T_E}{T_D} \uparrow,$$

the practical need for delegated local authority increases.

This does not mathematically prove independence. It predicts pressure toward:

- delegated emergency powers;
- local courts;
- conditional autonomy;
- charter government;
- agency discretion;
- local tax retention;
- independent rescue/security;
- later constitutional conflict.

The frontier does not become free because law disappears. It becomes politically volatile because law arrives late.

31. Causal Missing Status

Loom non-return creates a legal state more complex than ordinary disappearance.

A person may be:

- dead;
- stranded;
- alive beyond current contact;
- translated somewhere not yet recoverable;
- deliberately absent;
- causally delayed.

Estates, marriages, debts, corporate authority, licenses, insurance, and synthetic lineage claims cannot always wait for definitive proof.

Causal Missing Status is therefore an institutional response to epistemic delay.

It is an example of the setting's core method: a fictional physical constraint generates a legal category rather than a supernatural exception.

32. Synthetic persons

Synthetic persons must be separated from non-person automation.

A complete 2226 census should distinguish:

$$P_{\text{total}} = P_{\text{bio}} + P_{\text{synth}},$$

while machines without recognized personhood remain capital rather than population.

The current Atlas contains a promoted recognized-synthetic-person census of 214,068,429. Detailed individuation/staffing calibration remains provisional in some zones, especially heliocentric infrastructure.

Economically, synthetic persons can own, work, contract, inherit, migrate, and create institutions if recognized by the relevant jurisdiction.

This makes personhood law an economic variable, not merely a philosophical debate.

33. Security geography

Military and security presence follows:

- population;
- traffic;
- rescue burden;
- strategic resources;
- shipyards;
- sensor coverage;
- metric/Loom infrastructure;
- route centrality.

A low-population orbital node can therefore have greater security weight than a much larger settlement.

The same logic applies to piracy and organized crime. Criminal opportunity concentrates where high-value packets, stale jurisdiction, weak rescue, and chokepoint traffic overlap.

34. Route shock model

Suppose edge e is lost.

Define baseline shortest-path cost $d_G(i, j)$ and post-loss cost $d_{G-e}(i, j)$.

The route shock is

$$\Delta d_{ij}^{(e)} = d_{G-e}(i, j) - d_G(i, j).$$

Aggregate economic exposure can be approximated by

$$S_e = \sum_{i,j} F_{ij} \Delta d_{ij}^{(e)},$$

where F_{ij} is traffic or economic flow.

High- S_e edges are strategic even if they carry modest routine volume.

The downstream effects include:

route loss $\rightarrow$ latency $\uparrow \rightarrow$ insurance $\uparrow \rightarrow$ inventory $\uparrow \rightarrow$ local autonomy $\uparrow$.

This is a directly testable game/economic engine.

35. Mc production shock

Let qualified annual Mc demand be D_M and normal serviceable supply be S_M . CANON II v2.2 fixes the market state qualitatively as tight at the margin:

$$S_M \approx 1.2 \text{ t y}^{-1}, \quad D_M > S_M \text{ modestly at the margin.}$$

The exact demand tonnage is intentionally not canonized; the **sign of tightness** is.

A primary-production shock

$$\dot{M}_{\text{primary}} \downarrow$$

propagates as

commissioning/retrofit capacity $\downarrow \rightarrow$ fleet scarcity $\uparrow$
 $\rightarrow$ urgent mobility price $\uparrow$
 $\rightarrow$ route concentration $\uparrow$
 $\rightarrow$ frontier inventories/autonomy $\uparrow$

For a concrete sensitivity, a 20% shock to the $\sim 1.0 \text{ t/y}$ primary term reduces primary output to $\sim 0.8 \text{ t/y}$. Holding a central $\sim 0.2 \text{ t/y}$ recovered/requalified flow constant gives

$$S'_M \approx 1.0 \text{ t y}^{-1},$$

roughly a 17% fall in effective serviceable throughput from the $\sim 1.2 \text{ t/y}$ central baseline. Because the market already operates near capacity, the first-order consequences are queue lengthening, reprioritization, retrofit delay, reserve draw, and premium escalation rather than an abstract shock absorbed by unknown slack.

The response becomes nonlinear when governments or major institutions reserve material for military, survey, rescue, navigation, or infrastructure fleets.

36. No-FTL-comms counterfactual

A useful falsification exercise is to add independent FTL communication.

If arbitrary low-cost FTL communication exists, then:

- courier information value collapses;
- packet waiting time collapses;
- causal market arbitrage shrinks;
- remote command becomes much stronger;
- local delegated authority weakens;
- synchronization conflicts diminish;
- frontier legal isolation falls;
- the inherited courier's information-security role changes radically.

Therefore no-independent-FTL-comms is not decorative lore. It is a load-bearing civilizational parameter.

37. Loom-arrival counterfactual

Another useful test is to move practical Loom discovery from c.2220 to c.2120.

If Loom existed for a century, it would alter:

- Solar industrial investment;
- population distribution;
- military basing;
- port hierarchy;
- corporate concentration;
- migration;
- scientific institutions;
- constitutional law.

The current setting explicitly avoids this.

Loom is too recent to have **generated** the mature Solar System. This is a governing chronology constraint, not yet a complete econometric proof that Loom has zero 2221-2226 feedback on asset values or investment. The promoted census and mature industrial geography are inherited torch/early-metric stocks; the first-order Loom effects by 2226 should appear in marginal flows - investment, insurance, route speculation, specialist movement, inventory policy, and frontier institutions - rather than retroactively rebuilding centuries of settlement.

38. Sensitivity hierarchy

Paper III's highest-leverage uncertain variables are:

1. global/regional fertility;
2. longevity distribution and access;
3. habitat construction cost and commissioning rate;
4. automation/productivity;
5. ownership and redistribution institutions;
6. electricity/heat cost;
7. certification automation;
8. Mc production;
9. metric fleet penetration;
10. Loom fleet count;
11. route redundancy;
12. packet frequency;
13. route reliability;
14. insurance/legal enforceability;
15. synthetic-person individuation, migration, and rights calibration.

Not all uncertainty deserves equal modeling effort.

A one-ppm change in torch deposition may dominate a ship design; a one-point change in TFR can dominate centuries of demography. Paper III therefore ranks sensitivities by downstream consequence rather than narrative interest.

39. Research program

The next academically serious economic work should include:

39.1 Conserved cohort-detail demography

Extend the already-conserved Solar census into one sex/age/fertility/longevity/migration ledger while preserving frozen system totals and preventing cohort double-counting.

39.2 Habitat capital model

Estimate pressure volume, shielding, power, thermal rejection, transport, maintenance, and civic infrastructure per additional resident.

39.3 Synthetic-person individuation dynamics

Starting from the promoted 214,068,429 recognized-person census, model instantiation, legal recognition, fork/divergence/reconciliation, labor participation, staffing, and migration.

39.4 Input-output matrix

Construct a reduced 2226 A matrix for energy, bulk materials, precision materials, compute, habitats, ships, medicine, transport, certification, and services.

39.5 Route-flow engine

Combine Loom graph costs with packet frequency, reliability, fleet capacity, insurance, and shortest-path substitution.

39.6 Capital/ownership model

Derive labor share, machine-capital share, synthetic-person income, taxes, dividends, and household purchasing power rather than selecting them.

39.7 Port throughput model

Use the 127-node major/strategic Solar infrastructure graph to model traffic, investment, queueing, failure, and security allocation.

39.8 Certification economics

Determine which proof/metrology functions remain scarce after mature AI/robotic automation.

40. Hard kills

The civilizational model fails if it requires:

1. bulk interstellar trade to sustain the 2226 economy;
2. Loom to explain demographics that developed before Loom existed;
3. independent FTL communications while retaining packet civilization;
4. free or instantaneous terminal access despite metric stand-off;
5. population numbers that do not conserve people;
6. synthetic persons counted simultaneously as owned capital and legal persons;
7. universal lunar D-³He grid supply without a physically credible extraction chain;
8. post-scarcity claims that ignore fixed location, heat, capital, authority, or access;
9. megacorporate dominance without an actual concentration mechanism;
10. route chokepoints that do not follow from the graph;
11. market prices that know information which has not physically arrived;
12. remote governments exercising real-time knowledge or control without a causal channel.

41. Discussion

The most important economic consequence of LOOM physics is not faster trade.

It is **plural distance**.

A person in 2226 can be physically close but institutionally far; metrically distant but relationally adjacent; economically central but demographically tiny; legally governed by an authority whose latest instruction is weeks old; or sitting on information that is true but not yet true *for anyone else who can prove it*.

That creates a civilization in which information has mass because information requires a carrier.

It also creates a useful inversion of conventional space-opera economics. Energy is not the dominant universal scarcity. Bulk matter is not automatically the dominant scarcity. What becomes valuable are high-performance states of matter, trusted proof, synchronization, access, route knowledge, authority, location, and the right to make a decision before someone else's version of reality arrives.

The early interstellar frontier is therefore not a mass-colonization boom.

It is an epistemic, legal, scientific, commercial, and constitutional shock delivered by a tiny number of ships into an old civilization.

That is why a small courier matters.

It can carry almost no fraction of civilization's material output and still move things civilization cannot cheaply reproduce at the destination:

knowledge + authority + trust + people + unique matter.

42. Conclusion

Papers I–III now form one causal argument.

Paper I:

$M1+M2 \rightarrow \text{relational embedding physics.}$

Paper II:

relational physics $\rightarrow$ finite ships, materials, heat, metrology, certification.

Paper III:

engineering constraints $\rightarrow$ transport cost, network geography, scarcity, institutions.

The final civilizational spine is

ordinary physics + M1 + M2

$\Downarrow$

torch + metric + Loom + no independent FTL comms

$\Downarrow$

two geographies + strategic materials + packet latency

⇓

agglomeration + chokepoints + local authority + causal markets

⇓

the first five minutes of the interstellar age.

Humanity discovers that ordinary spacetime is not the deepest geography and immediately does what humanity always does with geography: it measures it, finances it, insures it, regulates it, monopolizes parts of it, fights over it, falls in love across it, gets lost in it, and argues about who has the right to decide what happens next.

That is the economic and institutional consequence of relational transport.

The red-team result is therefore deliberately asymmetric: the population ledger and core plural-distance/packet-civilization logic reproduce cleanly, while the generalized cost, ownership, route-flow, and concentration models remain less numerically mature than Paper II's ship arithmetic. Paper III v1.0 makes that status explicit rather than presenting plausible mechanisms as unique equilibria.

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4. Samuelson, P. A. (1954). *The Transfer Problem and Transport Costs, II: Analysis of Effects of Trade Impediments*. Economic Journal 64(254), 264–289.
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6. Schelling, T. C. (1978). *Micromotives and Macrobehavior*. W. W. Norton.
7. Canon demographic values are inherited from the LOOM 2226 canon baseline; non-canon energy, automation, labor, and price calibrations are project-model outputs and are not attributed to these references.

Appendix A — Current economic/demographic calibration card

Variable	Current value/status
Conserved biological humanity	8,571,824,600 - canon
Recognized synthetic persons	214,068,429 - canon
Earth share of biological humanity	~97.0% - derived from canon census
Biological humans off Earth	~259.3M - conserved within biological total
Major/strategic Solar infrastructure graph	127 nodes - canon
Primary flight-grade Mc production	~1,000 kg/y - canon
Recovered/requalified Mc flow	~150-250 kg/y - canon
Working central serviceable Mc flow	~1.2 t/y - canon calibration
Cumulative flight-grade Mc	~8.7 t - canon

Variable	Current value/status
Qualified Mc demand	modestly exceeds normal serviceable supply at margin - canon qualitative state
Continuous civilization electricity	~66.8 TW - Paper III working model output, non-canon
Mature-Earth delivered electricity	~\$0.025-0.055/kWh - provisional model
Machine-executable task-hours	~82-90%; working central ~86% - provisional model
Mature-Earth paid workweek	~20-24 h/week; central ~22 - provisional model
Labor income share	~35-42% - provisional model
Ordinary reproducible material/service basket	~28-35% of 2026 real cost - provisional model
Independent FTL communications	None - canon
Interstellar demographic effect by 2226	Loom too recent to drive mature Solar demography - canon constraint

Appendix B — Promoted conserved Solar census

System	Biological	Synthetic
Earth	8.313B	132.39M
Earth Orbit	64.74M	10.76M
Moon	14.96M	4.07M
Mars	60.29M	6.16M
Mercury	9.24M	6.53M
Venus	26.28M	1.78M
Ceres	7.73M	4.46M
Belt Majors	2.26M	2.22M
Jupiter	24.18M	9.94M
Saturn	31.27M	10.37M
Uranus	8.08M	5.99M
Neptune	6.84M	6.45M
Kuiper	3.22M	5.44M
Oort	242	~1.2k
Heliocentric	206k	7.50M

System totals are the promoted conserved authority. The heliocentric synthetic-person allocation remains specifically flagged for further individuation/staffing audit.

Appendix C — Standing reproducibility package for journal-facing release

A journal-facing release should be accompanied by code/data sufficient to reproduce:

- generalized transport-cost scenarios;
- reduced 2226 input-output matrix and Leontief inverse;
- Mc production shock propagation;
- Loom graph shortest paths, centrality, and route-loss shocks;
- packet-frequency/latency distributions;
- insurance risk pricing;
- conserved cohort-detail Solar-System demography;
- habitat-capital constraints;
- energy-sector reconstruction;
- relative-price basket;
- labor/capital ownership scenarios;

- port throughput and investment model.

Every exact-looking economic number must be a source, model output, or explicit calibration.